

Unit 54 Tests for functional groups, separation and purification of compounds

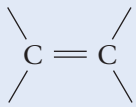
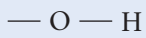
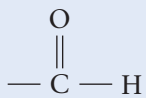
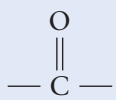
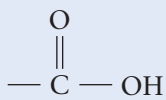
Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

54.1 Chemical tests for various functional groups in carbon compounds

We have studied chemical tests for identifying inorganic species in the last unit. Here we will discuss chemical tests for identifying various functional groups in carbon compounds.

We are going to look at chemical tests for functional groups of homologous series shown in Table 54.1.

Table 54.1 Functional groups of different homologous series

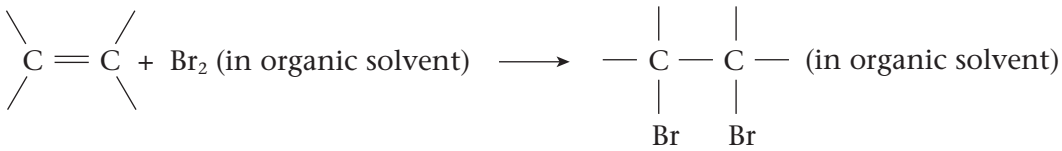
Homologous series	Functional group
Alkenes	
Alcohols	
Aldehydes	
Ketones	
Carboxylic acids	

54.2 Tests for the alkene functional group (–C=C–) N1

CE 2006 Paper 1 Q1(b)(iii) AL 2002 Paper 1 Q8(a)(ii)
ASL 2000 Q7(b) AL 2005 Paper 2 Q7(a)
ASL 2005 Q9(b) AL 2006 Paper 2 Q5(c)
AL 2000 Paper 1 Q8(a)(i)

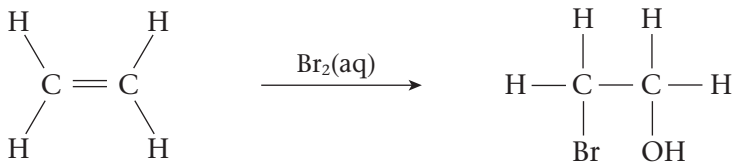
Bromine test N2

When bromine dissolved in an organic solvent is added to an alkene, the orange bromine solution becomes colourless rapidly at room temperature and in the absence of light. This is because an addition reaction occurs*.



Often the test for an alkene involves shaking the alkene with aqueous bromine rather than pure bromine.

In the case of ethene, 1,2-dibromoethane is obtained. This does not affect what we observe — the aqueous bromine is still decolourized.



2-bromoethanol



Fig. 54.1 Aqueous bromine is decolourized by ethene

* We have discussed addition reactions of alkenes in Topic 7 Fossil Fuels and Carbon Compounds and Topic 8 Chemistry of Carbon Compounds.

N1 Combustion test

Combustion test provides us a crude idea about the degree of saturation of a carbon compound.

During the test, heat a small amount of the compound under test in a non-luminous Bunsen flame in the fume cupboard, then remove it from the flame as it ignites. Observation of the flame colour helps us make deductions about the degree of unsaturation of the compound.

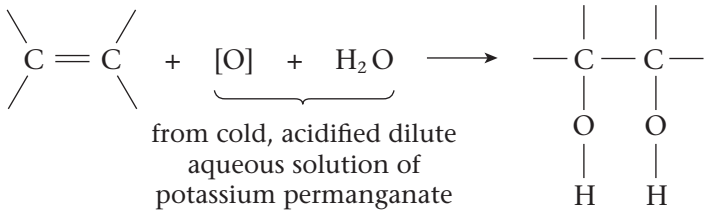
- Aliphatic saturated compounds tend to burn with blue or clear yellow flame.
- Unsaturated compounds burn with a smoky flame.

N2 Examination questions often ask about using bromine test for identifying alkenes.

Treatment with acidified aqueous solution of potassium permanganate

CE 2006 Paper 1 Q1(b)(iii)
CE 2010 Paper 2 Q38
CE 2011 Paper 2 Q50
ASL 2005 Q9(b)
AL 2002 Paper 1 Q8(a)(ii)

When cold, acidified dilute aqueous solution of potassium permanganate is added to an alkene, the purple solution becomes colourless quickly. This is because an addition reaction occurs.



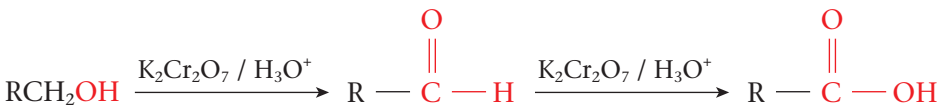
54.3 Tests for the alcohol functional group (–OH) N3

N3 Refer to page T41.

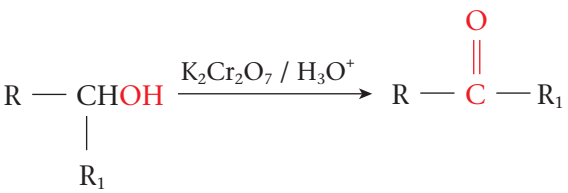
Oxidation by acidified aqueous solution of potassium dichromate

* We have discussed the oxidation of alcohols in Topic 8 Chemistry of Carbon Compounds.

Primary and secondary alcohols differ from tertiary alcohols in their reactions with acidified aqueous solution of potassium dichromate. Primary alcohols yield aldehydes, which are further oxidized to carboxylic acids. Secondary alcohols react to yield ketones*.



primary alcohol



secondary alcohol

N4 Examination questions often require students to distinguish a primary / secondary alcohol from a tertiary alcohol based on whether the alcohol can turn acidified potassium dichromate solution green.

DSE 2012 Paper 2 Q3(a)(i)
ASL 2002 Q12(a)(i)
AL 2000 Paper 1 Q8(a)(i)
AL 2005 Paper 2 Q7(a)
AL 2008 Paper 2 Q7(c)

During the test, add dilute sulphuric acid followed by aqueous solution of potassium dichromate to the sample under test. Warm the mixture and notice any colour change.

For primary and secondary alcohols, the clear orange dichromate solution becomes opaque and turns green almost immediately. Tertiary alcohols give negative test because they are difficult to oxidize. N4

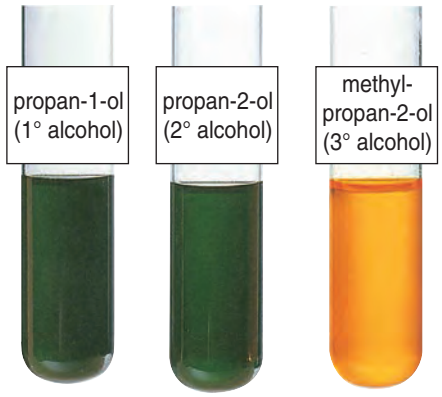


Fig. 54.2 When primary and secondary alcohols are treated with acidified aqueous solution of potassium dichromate, the orange dichromate solution turns green, but tertiary alcohols show no colour change

Extension

* We will discuss the test in more details under the test for carbonyl group.

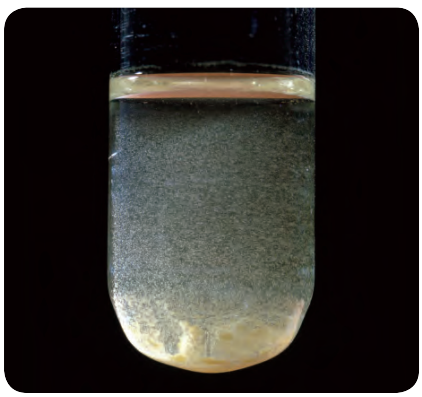
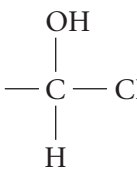


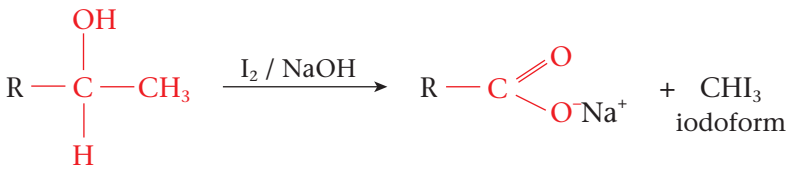
Fig. 54.3 The yellow precipitate of iodoform

Iodoform test

AL 2002 Paper 1 Q5(a)(ii)

The main use of iodoform test* is for the identification of ketones with a methyl group attached to the carbonyl group. However, ethanol and secondary alcohols with the  group also give a bright yellow precipitate of **iodoform** (i.e. triiodomethane, CHI_3).

During the test, the sample under test is warmed with iodine and an aqueous solution of sodium hydroxide.



Lucas reagent

AL 1999 Paper 2 Q7(a)

A solution of anhydrous zinc chloride in concentrated hydrochloric acid is known as the **Lucas reagent**.

When an alcohol is treated with the Lucas reagent, the corresponding chloroalkane is formed. Since chloroalkane is immiscible with water, the appearance of cloudiness in the aqueous phase indicates that a reaction has occurred.

$$\text{R–OH} \xrightarrow[ZnCl_2]{\text{concentrated HCl}} \text{R–Cl}$$

iodoform 碘仿　　Lucas reagent 盧卡斯試劑

Extension

* The order of reactivity of alcohols in this reaction is:
$3° > 2° > 1°$
We have discussed this in Topic 8 Chemistry of Carbon Compounds.

Primary, secondary, and tertiary alcohols differ from one another in their reactions with the Lucas reagent*.

$$(CH_3)_3COH \xrightarrow[ZnCl_2]{\text{concentrated HCl}} (CH_3)_3CCl$$

3° alcohol — cloudiness appears in 1 minute

$$(CH_3)_2CHOH \xrightarrow[ZnCl_2]{\text{concentrated HCl}} (CH_3)_2CHCl$$

2° alcohol — cloudiness appears in 5 minutes

$$CH_3CH_2OH \xrightarrow[ZnCl_2]{\text{concentrated HCl}} CH_3CH_2Cl$$

1° alcohol — cloudiness appears in hours

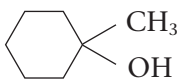
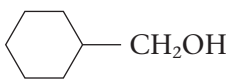


Fig. 54.4 Cloudiness appears in 1 minute for the reaction of a tertiary alcohol with the Lucas reagent

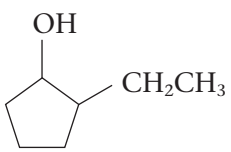
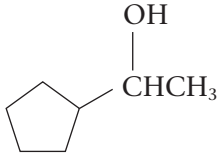


54.1 Distinguishing between two compounds in a pair

Suggest a chemical test to distinguish one compound from the other in each of the following pairs. Your answer should include the observations expected.

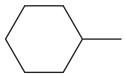
a)  and 

Extension

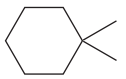
b)  and 

Solution

a) The first compound is a tertiary alcohol while the second one is a primary alcohol. The two compounds can be distinguished by warming with an acidified aqueous solution of potassium dichromate.



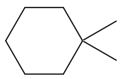
is oxidized and the orange dichromate solution changes to green.

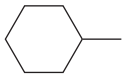


is not oxidized and shows no observable change.

Extension

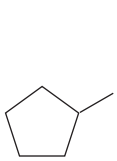
Alternatively, the two compounds can be distinguished by the Lucas reagent.

Cloudiness appears rapidly when 

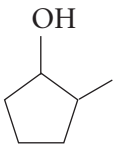
is treated with the Lucas reagent. There is no immediate observable change for 

.

b) The two compounds can be distinguished by warming with iodine and an aqueous solution of sodium hydroxide (i.e. iodoform test).



gives a yellow precipitate (CHI_3) while there is no observable change for



.

54.2 Deducing the structure of an alcohol from structures of products of its reactions

Alcohol A has a molecular formula of $C_5H_{12}O$. Reaction of A with an acidified aqueous solution of potassium dichromate produces a compound B, $C_5H_{10}O_2$. Heating A over Al_2O_3 produces 3-methylbut-1-ene. Suggest structures for A and B, and explain the reactions involved.

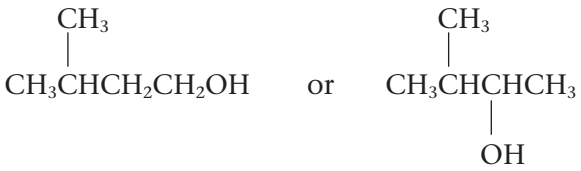
Solution

The structure of A is $CH_3CHCH_2CH_2OH$ and that of B is CH_3CHCH_2COOH.

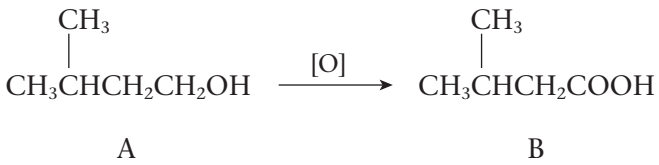
Dehydration occurs when A passes over Al_2O_3. The product is 3-methylbut-1-ene:

$CH_3CHCH{=}CH_2$

From the structure of this dehydration product, we can deduce that the structure of A is either



Oxidation of A to B gives an additional oxygen atom in the product B. This suggests that B is a carboxylic acid and hence A is a primary alcohol, i.e.



A B



Checkpoint

Refer to page T41.

1 Suggest a chemical test to distinguish one compound from the other in each of the following pairs. Your answer should include the reagents used and the observations expected.

a) propan-1-ol and propene

Extension

b) propan-1-ol and propan-2-ol

c) pentan-2-ol and pentan-3-ol

2 Pentan-1-ol and pentan-2-one can be distinguished from each other by using acidified aqueous solution of potassium permanganate. Describe the chemistry involved.

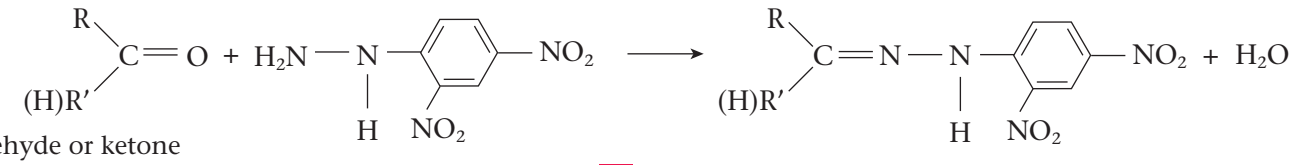
54.4 Tests for the aldehyde and ketone functional groups (–CHO and >C=O)

Treatment with 2,4-dinitrophenylhydrazine

ASL 2011 Paper 1 Q7(b)
ASL 2012 Paper 2 Q6(a)

Aldehydes and ketones react with **2,4-dinitrophenylhydrazine** to form yellow to red precipitate of 2,4-dinitrophenylhydrazones. This reaction is used to test for the presence of carbonyl group  in samples.

2,4-dinitrophenylhydrazine 2,4-二硝基苯肼



aldehyde or ketone

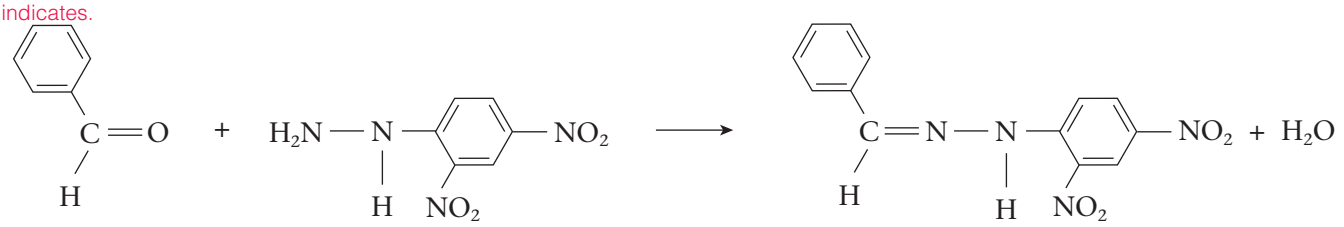
2,4-dinitrophenylhydrazine N5

2,4-dinitrophenylhydrazone

N5 Students should be able to give the correct structure of 2,4-dinitrophenylhydrazine.

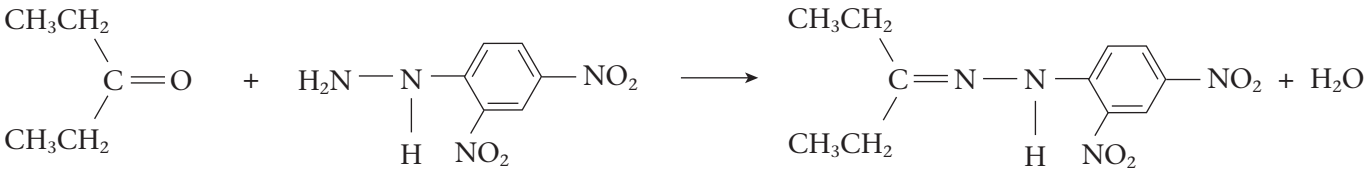
Examination questions often ask about what functional group the 2,4-dinitrophenylhydrazine test indicates.

Benzaldehyde and pentan-3-one react with 2,4-dinitrophenylhydrazine according to the following equations:



benzaldehyde

benzaldehyde 2,4-dinitrophenylhydrazone



pentan-3-one

pentan-3-one 2,4-dinitrophenylhydrazone

DSE 2012 Paper 2 Q3(a)(ii) ASL 2010 Paper 1 Q5(b) AL 2002 Paper 1 Q5(a)(ii)
DSE 2015 Paper 2 Q3(b)(i) AL 2001 Paper 1 Q8(b)(iii) AL 2008 Paper 2 Q7(c)
ASL 2005 Q9(b)

During the test, add 2,4-dinitrophenylhydrazine test reagent to the sample. Formation of a yellow to red precipitate indicates the presence of a carbonyl group.

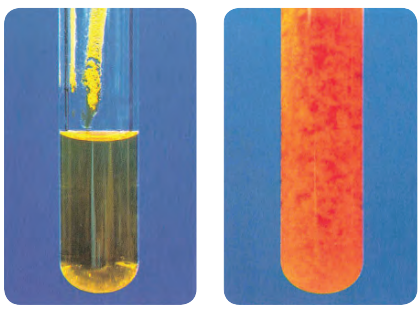


Fig. 54.5 Ethanal forms an orange precipitate with 2,4-dinitrophenylhydrazine

A positive 2,4-dinitrophenylhydrazine test does not distinguish between an aldehyde and a ketone. This distinction can be made by observing the oxidation of the carbonyl compound with acidified aqueous solution of potassium dichromate. Aldehydes undergo oxidation very easily, whereas ketones are difficult to oxidize.

DSE 2012 Paper 2 Q3(a)(i) ASL 2012 Paper 2 Q6(b)
DSE 2015 Paper 2 Q3(b)(ii) AL 2012 Paper 2 Q9(b)

Treatment with Tollens' reagent (silver mirror test) N6

N6 Refer to page T42.

Tollens' reagent is an aqueous solution of silver nitrate in ammonia. The reagent contains the diamminesilver(I) ions, $[Ag(NH_3)_2]^+$. Although this ion is a very weak oxidizing agent, it oxidizes aldehydes to carboxylate ions. In the reaction, $[Ag(NH_3)_2]^+$ ions are reduced to metal silver which deposits on the wall of the reaction vessel as a silver mirror*.

DSE 2014 Paper 2 Q3(c)(ii)(I)

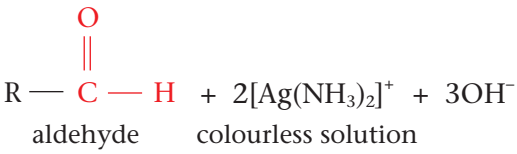
* If the wall of the reaction vessel is not clean enough, a silver mirror cannot form and a grey or black precipitate is deposited instead.

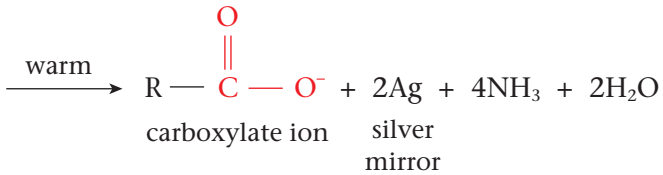
Video
Testing ethanal with Tollens' reagent

Tollens' reagent 托倫斯試劑

ASL 2002 Q7(I)(b)

In the Tollens' test (also known as the silver mirror test), aldehydes are mixed with the Tollens' reagent in a clean test tube and placed in a water bath kept at 60 °C*.





ASL 2012 Paper 2 Q6(b)
AL 2003 Paper 2 Q7(c)
AL 2012 Paper 2 Q9(b)

Tollens' reagent gives a negative result with all ketones. Therefore we can use it to distinguish aldehydes from ketones.

* There is an explosion hazard if the mixture is heated to dryness.



Fig. 54.6 An aldehyde gives silver deposits or a silver mirror with the Tollens' reagent

Extension

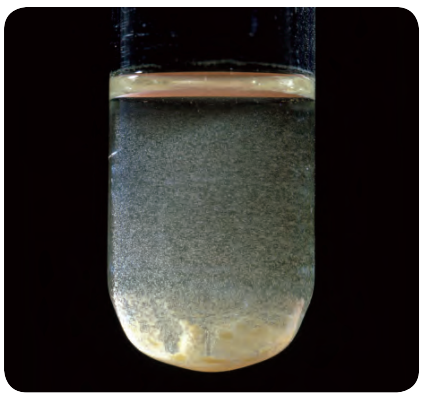


Fig. 54.7 The yellow precipitate of iodoform

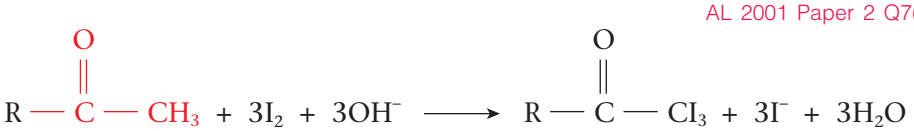
Extension

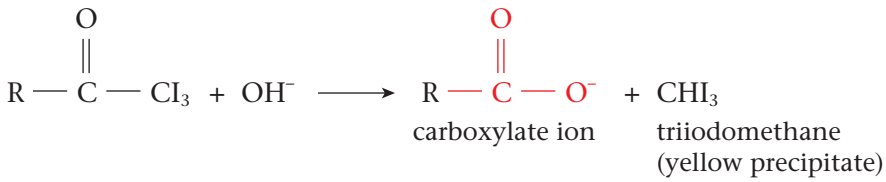
AL 2000 Paper 1 Q8(a)(i)

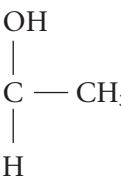
Triiodomethane formation (iodoform test)

Aldehydes and ketones having the  group react with iodine in an aqueous solution of sodium hydroxide to give a bright yellow precipitate of iodoform (i.e. triiodomethane, CHI_3).

AL 2001 Paper 2 Q7(b)(i)





Ethanol and secondary alcohols with the  group also give a positive result in the iodoform test. In the reaction, they are first oxidized to the  group and further reactions give the carboxylate ion and the iodoform.

N7 Refer to page T43.

Table 54.2 summarizes the common reagents used for testing aldehyde and ketone functional groups and the observations of the reactions. N7

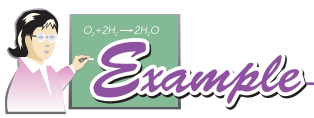
Table 54.2 Common reagents used for testing aldehyde and ketone functional groups and the observations of the reactions

Reagent	Aldehyde functional group $-\overset{\overset{O}{\|}}{C}-H$	Ketone functional group $-\overset{\overset{O}{\|}}{C}-$
2,4-dinitrophenylhydrazine	a yellow to red precipitate forms	
Warm with Tollens' reagent	silver mirror forms on the wall of reaction vessel	no reaction
Extension Iodoform test (test for $-\overset{\overset{O}{\|}}{C}-CH_3$ group)	a yellow precipitate of CHI_3 forms	

In the following activity, you are going to study the reactions of an aldehyde and a ketone with the reagents discussed above.

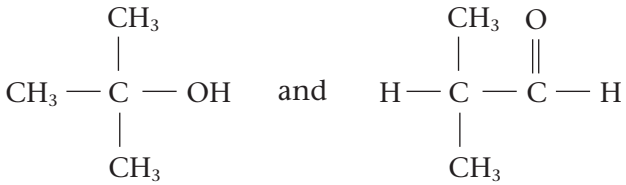
Activity 54.1

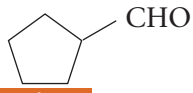
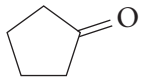
Studying the reactions of an aldehyde and a ketone with some reagents.



54.3 Distinguishing between two compounds in a pair

Suggest a chemical test to distinguish one compound from the other in each of the following pairs. Your answer should include the reagents used and the observations expected.

a) 

b)  and 

Extension

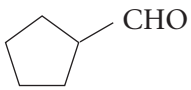
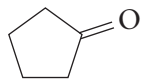
c) $CH_3CH_2-\overset{\overset{O}{\|}}{C}-CH_2CH_3$ and $CH_3CH_2CH_2-\overset{\overset{O}{\|}}{C}-CH_3$

Solution

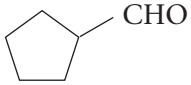
a) Treat the compounds with 2,4-dinitrophenylhydrazine separately.

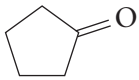
There is no observable change with $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-OH$ while $H-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-\overset{\overset{O}{\|}}{C}-H$ gives a yellow to red precipitate.

b) Distinguish the two compounds by warming with the Tollens' reagent.

 gives a silver mirror while there is no observable change for  .

Alternatively, the two compounds can be distinguished by warming with acidified aqueous solution of potassium dichromate.

 can be oxidized and the orange dichromate solution turns green.

 is not oxidized and there is no observable change.

Extension

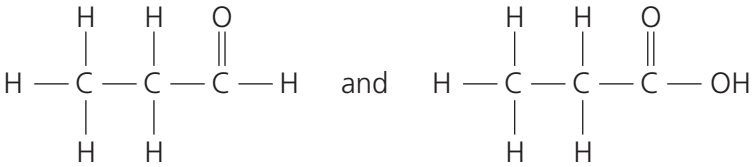
c) Distinguish the two compounds by warming with iodine and an aqueous solution of sodium hydroxide (i.e. the iodoform test).

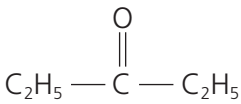
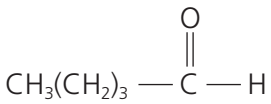
$CH_3CH_2-\overset{\overset{O}{\|}}{C}-CH_2CH_3$ shows no observable change while $CH_3CH_2CH_2-\overset{\overset{O}{\|}}{C}-CH_3$ gives a yellow precipitate.

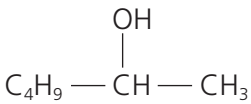
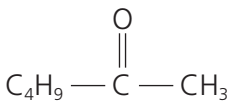
Checkpoint

Refer to page T43.

1 Suggest a chemical test to distinguish one compound from the other in each of the following pairs. Your answer should include the reagents used and the observations expected.

a)  and

b)  and 

c)  and 

2 Three carbon compounds, A, B and C, have the molecular formula C_4H_8O. All three compounds give an orange precipitate with 2,4-dinitrophenylhydrazine.

B and C also give a silver mirror when warmed with the Tollens' reagent.

a) What can be deduced about the three compounds?

b) Draw the structural formulae of A, B and C.

54.5 Tests for the carboxylic acid functional group (–COOH)

Treatment with aqueous solution of sodium hydrogencarbonate

CE 2010 Paper 2 Q38
AL 2002 Paper 1 Q5(a)(ii)

Carboxylic acids react with aqueous solution of sodium hydrogencarbonate to form sodium carboxylates, carbon dioxide and water.

$$RCOOH + NaHCO_3 \longrightarrow RCOO^-Na^+ + CO_2 + H_2O$$

Carboxylic acids are the most acidic carbon compounds. Other carbon compounds do not react with aqueous solution of sodium hydrogencarbonate. N8

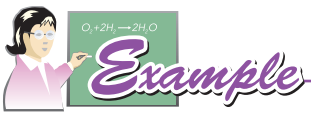
N8 Refer to page T44.

Ester formation

Warm the sample under test with ethanol and concentrated sulphuric acid. Then cool and pour the mixture into an aqueous solution of sodium carbonate. A carboxylic acid reacts to give a sweet, fruity-smelling ester that can be detected immediately.

$$RCOOH + CH_3CH_2OH \xrightarrow{H^+} RCOOCH_2CH_3 + H_2O$$

ester with a sweet, fruity smell



54.4 Distinguishing four liquids from one another

You are provided with four unlabelled bottles each containing one of the following colourless liquids:

propanone, propanoic acid, hexane, hex-1-ene

Outline a scheme of tests to distinguish the four liquids from one another. State also the expected observations of the tests.

Solution

We can distinguish the four liquids using the following steps:

1 Add water to the liquids. Only propanone and propanoic acid can mix with water.

2 Test the solutions obtained with pH paper. The solution of propanoic acid is acidic while that of propanone is not.

3 Add aqueous bromine to the other two compounds. Only hex-1-ene can decolourize the aqueous bromine rapidly.

Extension

54.5 Deducing the structures of compounds based on their reactions

A compound X ($C_8H_{10}O$) reacts with ethanoic acid in the presence of concentrated sulphuric acid to give a sweet-smelling liquid Y. Oxidation of X by dichromate ion yields Z (C_8H_8O). Z reacts with iodine and aqueous solution of sodium hydroxide to produce a yellow precipitate and sodium benzoate.

Deduce the structures of X, Y and Z. Explain your reasoning.

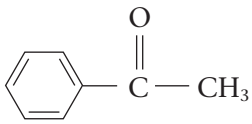
Extension

Solution

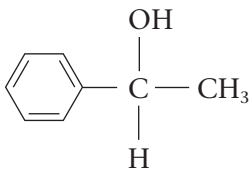
As X reacts with ethanoic acid to give a sweet-smelling liquid Y, we can deduce that X is an alcohol and Y is an ester.

As Z is an oxidation product of X and gives a positive result in the iodoform test, we can deduce that Z is a ketone with the  group.

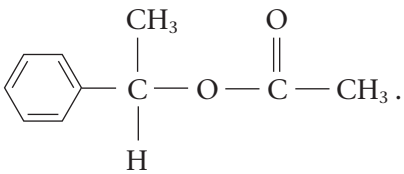
As a benzoate ion  is produced in the iodoform test, Z must have the following structure:



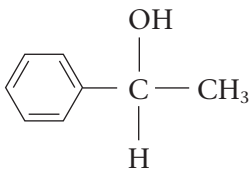
As Z is obtained by the oxidation of X, thus X is a secondary alcohol with the following structure:

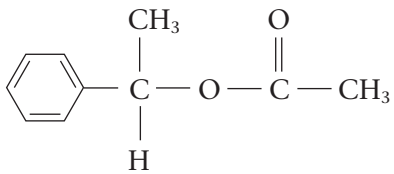


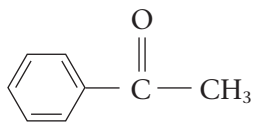
Y is an ester formed when X reacts with ethanoic acid, thus the structure of Y is

 .

To summarize, structures of X, Y and Z are as follows:







Checkpoint

In order to establish the functional groups present in an *acyclic* compound F (C_4H_8O), chemical tests were carried out. The test results are listed in the table below.

a) Complete the table by giving the deduction you can make from the result of each test.

Test reagent	Test result	Deduction
$K_2Cr_2O_7$ / H_3O^+	positive	functional group(s) that may be present: —OH, —CHO
2,4-dinitrophenylhydrazine	negative	functional group(s) absent: —CHO, $>C=O$
Aqueous bromine	positive	functional group(s) that may be present: $-C=C-$, $-C\equiv C-$

AL 2000 Paper 1 Q8(a)(i)

AL 2000 Paper 1 Q8(a)(i)

b) Based on the above information, draw three possible structures of F.

Any three of the following:
$H_2C=CHCH_2CH_2OH$
$H_3CCH=CHCH_2OH$
$H_2C=CHCH(OH)CH_3$
$H_3CCH=C(OH)CH_3$

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Problem Solving

You are provided with five unknown liquids labelled A to E.

They are known to be:

- ethanal (solution)
- ethanol
- propanone
- ethanoic acid (glacial)
- pentyl ethanoate

but NOT in that order.

Devise a scheme of chemical tests and use the results of the tests to identify each of the liquids.

Treat each liquid with the Tollens' reagent.
Only ethanal gives a silver mirror.
Treat the remaining four liquids with 2,4-dinitrophenylhydrazine.
Only propanone gives a yellow to red precipitate.
Mix the remaining three liquids with an aqueous solution of sodium hydrogencarbonate.
Only ethanoic acid gives effervescence.
Mix the remaining two liquids with phosphorus pentachloride.
Only ethanol gives steamy fumes.
The remaining liquid is pentyl ethanoate.

acyclic 無環的

54.6 Identifying functional group(s) in an unknown sample in the laboratory

After studying reagents for identifying functional groups in different homologous series, you are going to carry out chemical tests in the laboratory and identify the functional group(s) in a sample.



54.6 Identifying functional groups in a sample

Some experiments are carried out on a compound X and the results are shown below.

a) Complete the 'Deduction' column based on the results of the tests.

b) It is known that X does not contain any triple bond. Deduce, with explanation, the functional group(s) present in X.

Solution

Test	Procedure	Observation	Deduction
1	Ignite a small amount of X.	X burns with a sooty flame.	X has a high carbon content.
2	Dissolve a small amount of X in about 10 cm^3 of hot water.	A colourless solution forms.	X is polar or it can form hydrogen bonds with water.
3	Test the aqueous solution of X with blue litmus paper.	The solution turns the blue litmus red.	X is an acidic compound.
4	Add aqueous bromine dropwise to the aqueous solution of X and shake.	The aqueous bromine is decolorized rapidly.	X is unsaturated.
5	a) Add a spatula measure of X to about 2 cm^3 of ethanol; add a few drops of concentrated sulphuric acid and warm. b) Pour the contents of test tube into cold water in a beaker.	A sweet smell is detected.	Ester is formed. Therefore X contains the –COOH functional group.

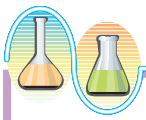
From the observation in Test 4, we can deduce that X is unsaturated. Since X does not contain any triple bond, it must contain the –C=C– functional group.

From the observation in Test 5, we can deduce that X must contain the –COOH functional group as it reacts with ethanol to give an ester.

Observations in Tests 2 and 3 agree with the above deduction as X dissolves in water to give an acidic solution.

Therefore X contains the –C=C– and –COOH functional groups.

In the following activity, you are going to identify the functional groups in two unknown compounds.



Activity 54.2

Identifying the functional groups in two unknown compounds.

54.7 Separation and purification of compounds

The preparation of a compound involves five main stages*:

- Stage 1 — planning the preparation;
- Stage 2 — carrying out the reaction to produce the desired product;
- Stage 3 — separating the crude product from the reaction mixture;
- Stage 4 — purifying and drying the product;
- Stage 5 — checking the purity and yield of the product.

In the following sections, we will discuss how to separate the crude product from a mixture of unreacted reagents and by-products and then purify the crude product. Finally, we will see how we can check the purity of the product.

* We have discussed this briefly in Topic 8 Chemistry of Carbon Compounds.

Table 54.3 shows the common separation and purification methods we usually employ for a liquid product and a solid product.

Table 54.3 Common separation and purification methods for products

Type of product	Separation and purification method to employ
Liquid product	• distillation • fractional distillation • liquid-liquid extraction
Solid product	• re-crystallization • chromatography

54.8 Distillation

During the preparation of a liquid product, we need to separate the crude product from a mixture of unreacted reagents and by-products. If these substances are non-volatile, we can separate the crude liquid product by **distillation** (Fig. 54.8).

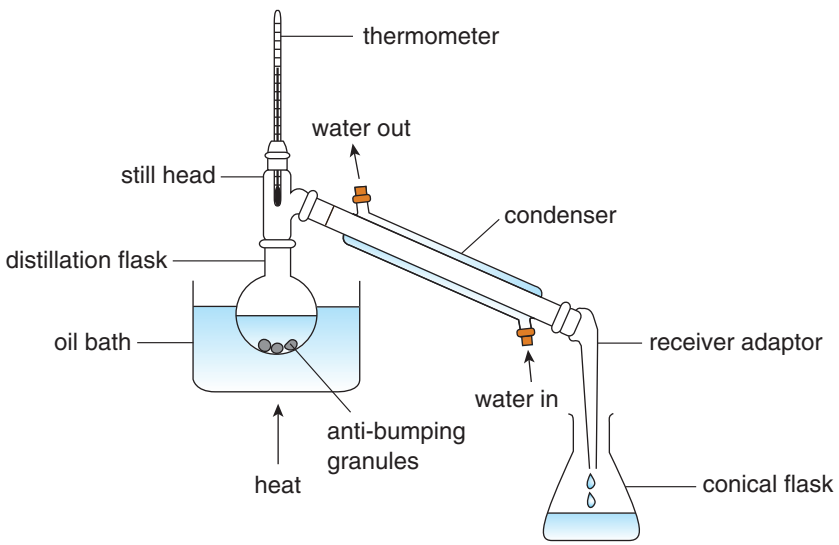


Fig. 54.8 A distillation set-up using an oil bath*

AL 2002 Paper 1 Q8(a)(i)
AL 2004 Paper 1 Q8(b)

* For distillation of low boiling flammable liquids, we usually use a water bath for heating. Otherwise, we may use an oil bath or a Bunsen flame.

distillation 蒸餾法

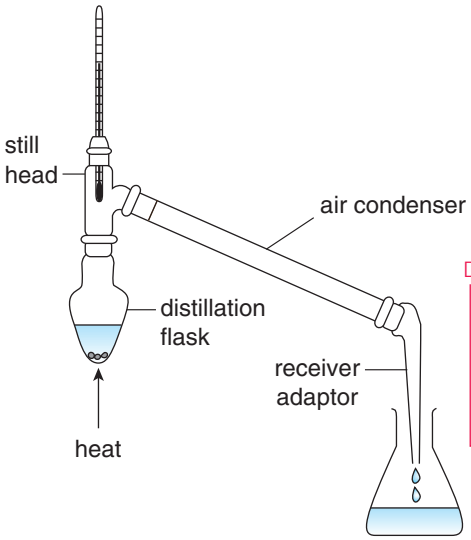


Fig. 54.9 Set-up for distillation with an air condenser

Use a water condenser for cooling compounds with boiling points up to about 140 °C. Cooling higher boiling compounds with a water condenser may boil the water passing through the condenser. This decreases the cooling efficiency and may crack the condenser. For compounds with boiling points higher than 140 °C, use an **air condenser** (Fig. 54.9).

DSE 2013 Paper 2 Q3(a)

54.9 Fractional distillation

We can separate a mixture of miscible liquids by **fractional distillation** providing that the boiling points of the liquids are not too close.

To separate a mixture of miscible liquids by fractional distillation, use the apparatus shown in Fig. 54.10. The **fractionating column** is packed with glass beads. This provides a larger surface area for condensation and vaporization of the liquid mixture.

AL 2002 Paper 1 Q8(a)(iii)

Reference Website

Refer to the following website for an animation explaining the principle of fractional distillation:
http://chem-ilp.net/labTechniques/FractionalDistillationlAnimation.htm
(accessed May 2015)

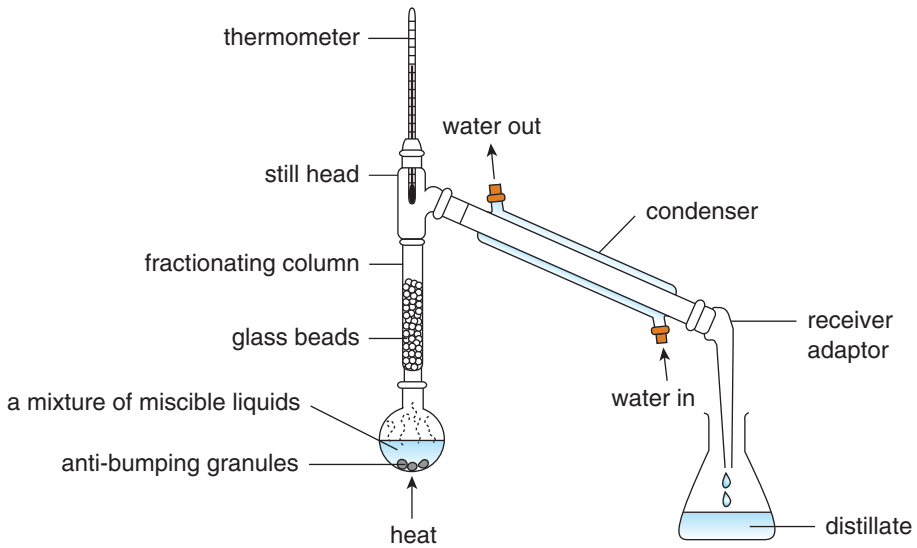


Fig. 54.10 Experimental set-up for fractional distillation of a mixture of miscible liquids

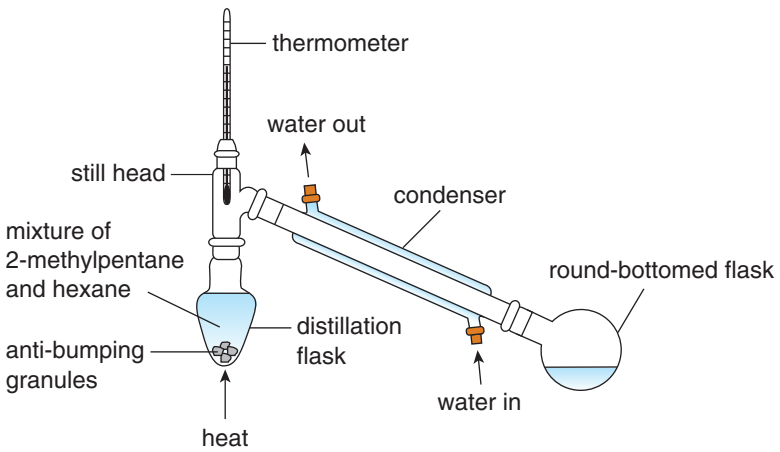
In general, for fractional distillation to work best, the difference in boiling points of liquids in the mixture should be larger than 10 °C. Otherwise, the separation is incomplete.

air condenser 空氣冷凝器 fractional distillation 分餾法 fractionating column 分餾柱

Example

54.7 Separating 2-methylpentane from a mixture of 2-methylpentane and hexane

A student suggests using the set-up shown below to separate 2-methylpentane from a mixture of 2-methylpentane (boiling point = 60 °C) and hexane (boiling point = 69 °C).



a) It is suggested that TWO modifications should be made to the set-up so as to carry out the experiment safely.

   i) The first modification is to fill the distillation flask to not more than three-fifths of its capacity. Explain why this modification is needed.

   ii) Suggest and explain another modification that should be made.

b) The set-up is then modified as suggested in (a). Explain whether the modified set-up can be used to separate 2-methylpentane from hexane effectively.

CE 2006 Paper 1 Q1(b)(i),(ii)

Solution

a) i) Otherwise, the liquid in the distillation flask may splash over into the condenser easily.

   ii) Pressure would build up inside the set-up when the mixture is heated. An explosion may happen. It is dangerous to conduct an experiment using a closed system.

   Modify the set-up by adding a receiver adaptor between the condenser and the round-bottomed flask.

b) No. The boiling points of 2-methylpentane and hexane are too close and they cannot be separated effectively by simple distillation.

Reference Website

Refer to the following website for an animation and a video clip on liquid-liquid extraction:
http://chem-ilp.net/labTechniques/SolventExtraction.htm
(accessed May 2015)

54.10 Liquid-liquid extraction

In the preparation of carbon compounds, it frequently happens that the liquid product is obtained in an aqueous solution together with ionic impurities. We can use the **liquid-liquid extraction** method to extract the product.

During the extraction, place the aqueous solution containing the product in a **separating funnel**. Then add an organic solvent that is immiscible with water, such as ethoxyethane* (Fig. 54.11).

N9

DSE 2013 Paper 2 Q3(a)

* Ethoxyethane is highly flammable. Carry out the extraction inside a fume cupboard.

AL 2003 Paper 2 Q5(a)(v)

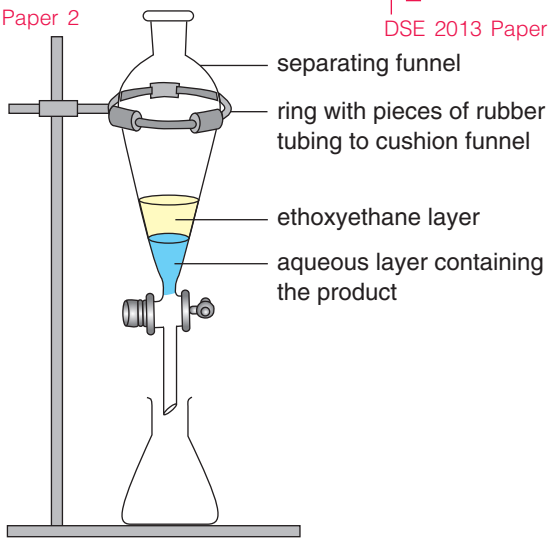


Fig. 54.11 During the extraction, add ethoxyethane to the aqueous solution containing the carbon compound product

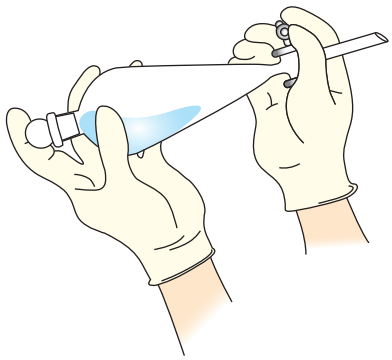


Fig. 54.12 Hold the stopper and tap firmly when shaking the separating funnel

N9 Refer to page T44.

Upon shaking the liquid mixture in the separating funnel, the product will dissolve preferentially in the organic solvent.

Hold the stopper and tap firmly when shaking the separating funnel* (Fig. 54.12). After shaking, allow the two layers to separate and run off the lower aqueous layer.

* The boiling point of ethoxyethane is 35 °C. During shaking, open the tap of the separating funnel regularly to release the pressure due to the ethoxyethane vapour caused by shaking and the heat of the hands.

We can re-extract the aqueous layer using fresh samples of ethoxyethane solvent so as to obtain more of the product.

The final stage of the process involves the following steps:

- collect together the different samples of ethoxyethane;
- remove the final traces of water using a drying agent such as anhydrous sodium sulphate.

DSE 2014 Paper 2 Q3(c)(i) AL 2007 Paper 1 Q8(d)
AL 2003 Paper 2 Q5(a)(iv) AL 2009 Paper 1 Q8(e)

- filter off the drying agent;
- obtain the product by distilling off the ethoxyethane solvent*.

DSE 2014 Paper 2 Q3(c)(i)

* Use a water bath when distilling off the ethoxyethane; use an ice-cooled receiver flask to receive the ethoxyethane.

AL 2003 Paper 2 Q5(a)(v)

liquid-liquid extraction 液液萃取　　separating funnel 分液漏斗



54.8 Isolating pure heptanoic acid from a mixture of two liquids based on liquid-liquid extraction

You are provided with a mixture of two liquids, heptanoic acid and cyclohexanone. Outline an experimental procedure, based on liquid-liquid extraction, to isolate pure heptanoic acid from the mixture.

Solution

AL 2012 Paper 1 Q8(a)(ii)

Add ethoxyethane and dilute sodium hydroxide solution to the mixture in a separating funnel and shake. The heptanoic acid reacts with sodium hydroxide to give sodium heptanoate.

Allow the organic layer and the aqueous layer to separate after shaking. The organic layer contains cyclohexanone while the aqueous layer contains sodium heptanoate.

Run off the aqueous layer. Regenerate the heptanoic acid by adding dilute hydrochloric acid to the aqueous layer. Then extract the heptanoic acid with ethoxyethane, remove final traces of water using a drying agent, and filter off the drying agent. Finally distil off the ethoxyethane to obtain pure heptanoic acid. N10

N10 Refer to page T45.

54.11 Re-crystallization

The **re-crystallization** method is often used to purify crude solid products.

We use this technique when the solid product dissolves readily in a hot solvent but is much less soluble when the solvent cools down.

A re-crystallization process involves three main stages: N11

N11 Examination questions may ask students to outline the re-crystallization procedure for the purification of a crude solid product.

ASL 2010 Paper 1 Q10(b)
AL 2001 Paper 1 Q8(a)(i)(II)
AL 2004 Paper 1 Q8(a)
AL 2007 Paper 2 Q7(a)(iv)
AL 2010 Paper 1 Q8(e)

Stage 1	Choose a suitable solvent which would not react with the solid product.
Stage 2	Use the minimum amount of hot solvent to dissolve the crude solid product. Filter to remove the insoluble impurities.
Stage 3	Allow the filtrate to cool slowly for crystals to re-form while soluble impurities stay in the cold solvent. Collect the crystals by filtration under reduced pressure and dry them.

re-crystallization 再結晶

Reference Website
Refer to the following website for an animation and a video clip on re-crystallization:
http://chem-ilp.net/labTechniques/Recrystallisation.htm
(accessed May 2015)

N12 Refer to page T45.

Stage 1 Choosing a suitable solvent

Choose a solvent that will dissolve the solid product while hot, but does not dissolve it while cold. Table 54.4 shows some common solvents for re-crystallization. N12

AL 2007 Paper 1 Q8(e)

Table 54.4 Some common solvents for re-crystallization

Solvent	Boiling point (°C)	Remark
Water	100	to be used whenever possible
Ethanol	78	flammable
Acetone (propanone)	56	flammable
Ethoxyethane	35	flammable, avoid whenever possible
Ethyl ethanoate	78	flammable

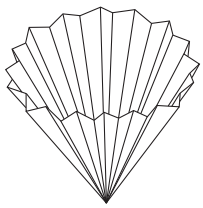
ASL 2010 Paper 1 Q10(b)
AL 2001 Paper 1 Q8(a)(i)(II)
AL 2004 Paper 1 Q8(a)
AL 2010 Paper 1 Q8(e)

Stage 2 Dissolving the crude solid product in the chosen solvent

Use the minimum amount of hot solvent to dissolve the solid product.

If coloured impurities are present, add a little activated carbon to remove them. After adding the activated carbon, heat the solution for 5 to 10 minutes. Filter the activated carbon and any insoluble impurities off using a piece of *fluted filter paper** inside a hot* stemless funnel* (Fig. 54.13).

* A piece of fluted filter paper (as shown below) is desirable when a rapid filtration is necessary, as in this re-crystallization process.



* Heating the set-up before filtration prevents crystals from forming on the paper and in the funnel.

* A funnel with a stem is prone to premature re-crystallization inside the stem because the filtrate can cool as it passes through the stem.

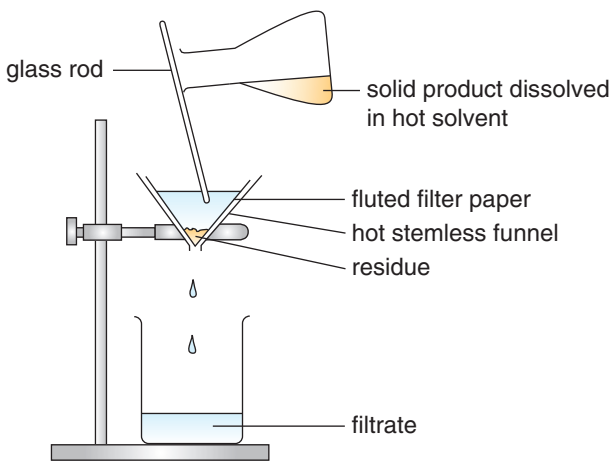


Fig. 54.13 Removing insoluble impurities by filtration

CE 2002 Paper 1 Q7(a)(ii)

fluted filter paper 槽狀濾紙

Reference Website
Refer to the following website for a video clip on filtration under reduced pressure:
http://chem-ilp.net/labTechniques/BuchnerFiltrationVideo.htm
(accessed May 2015)

Stage 3 Allowing crystals to form and collecting crystals by filtration under reduced pressure

Allow the filtrate to cool slowly to room temperature. Gradual cooling leads to the formation of large crystals.

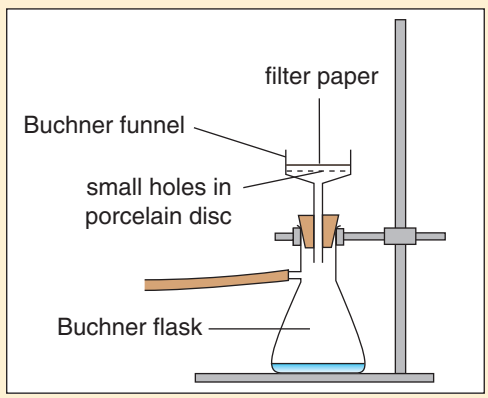
We often use filtration under reduced pressure when we need to separate solid crystals from a solution.

Extension

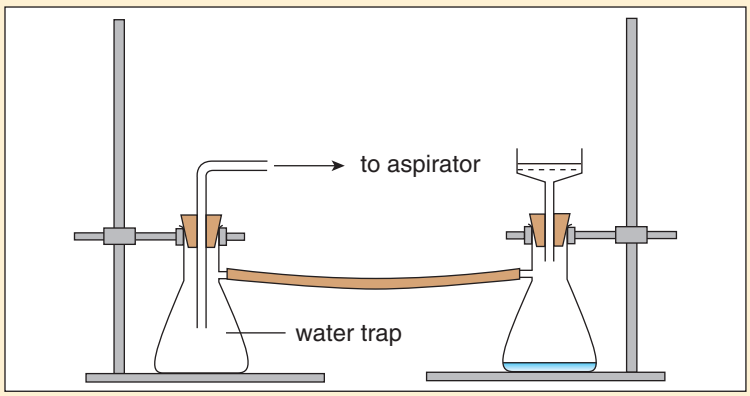
The following steps are involved:

- Use a **Buchner flask and funnel** in the process. Put a piece of filter paper in the funnel and wet the paper using a small amount of the solvent. Put the funnel in the flask and connect the rubber tubing (Fig. 54.14a).

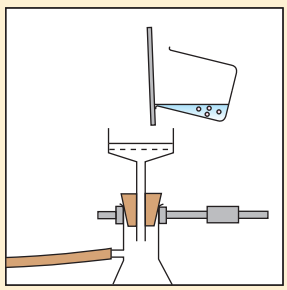
ASL 2010 Paper 1 Q10(a)
AL 2010 Paper 1 Q8(d)



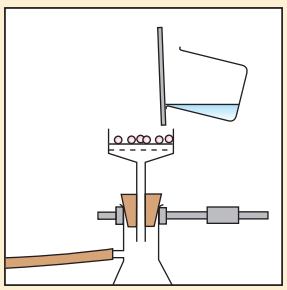
a) A Buchner flask and funnel connected with rubber tubing



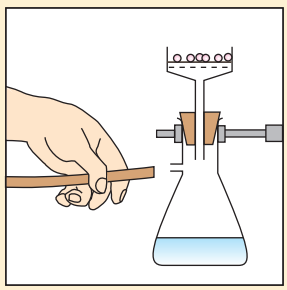
b) Set-up for filtration under reduced pressure



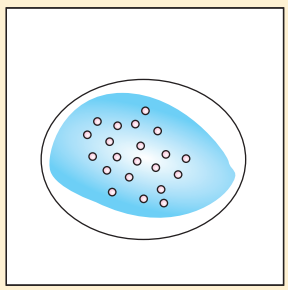
c) Pouring the crystals and the solution onto the filter paper



d) Washing crystals in the funnel with a small amount of the cold solvent



e) Stopping the suction by disconnecting the rubber tubing



f) Crystals in the funnel transferred to a watch glass

Fig. 54.14 Collecting crystals by filtration under reduced pressure

Buchner flask and funnel 布赫勒爾燒瓶及漏斗

Extension

- Connect the flask to a water trap, which is in turn connected to an **aspirator** (Fig. 54.14b).
- Pour the crystals and the solution onto the filter paper (Fig. 54.14c). The solution is pulled through the funnel rapidly.
- Wash the crystals with a small amount of the cold solvent (Fig. 54.14d).

Extension

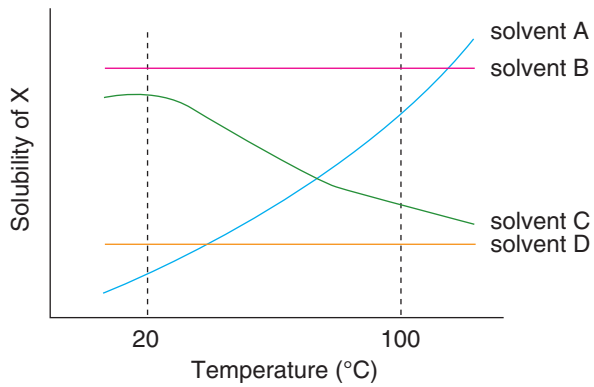
- When the crystals have been sucked dry as much as possible, stop the suction by disconnecting the rubber tubing* (Fig. 54.14e).
- Transfer the crystals to a watch glass (Fig. 54.14f). Dry the crystals by pressing them between filter papers.

Extension

* Disconnect the rubber tubing before turning off the aspirator. Otherwise, water may be sucked back into the Buchner flask.

Checkpoint

The graph below shows the solubility curves of a solid carbon compound X in four mutually miscible solvents A, B, C and D.



Which of the solvent is the best for the re-crystallization of X? Explain your answer.

Solvent A

The solubility of X in A is high at high temperatures but low at low temperatures.

AL 2001 Paper 1 Q8(a)(i)(I)

aspirator 吸氣器

54.12 Chromatography

We may employ **chromatography** to separate a complex mixture of substances that cannot be separated by the re-crystallization method readily.

The basic principle of chromatography is that the components in a mixture have different tendencies to *adsorb* onto a surface or dissolve in a solvent. There are several different types of chromatography, such as

- paper chromatography;
- column chromatography;
- thin layer chromatography (TLC);
- gas chromatography (GC);
- liquid chromatography (LC);
- high performance liquid chromatography (HPLC), etc.

All chromatographic methods require two different phases*:

- a **stationary phase** which remains stationary;
- a **mobile phase** which moves over or through the stationary phase. N13 N14

In this unit, we are going to consider the following three chromatographic methods:

- paper chromatography;
- column chromatography; and
- thin layer chromatography.

N13 Refer to page T46.

N14 Refer to page T47.

* A phase is defined as a uniform portion of matter separated from other portions of matter by a boundary surface. For example, a mixture of gases is a single phase because there is no boundary surface between the different gases. However, oil and water have a boundary surface and there are two phases.

54.13 Paper chromatography

This method is suitable for separating small amount of components in a mixture, and for finding out what these components are.

In **paper chromatography***, the stationary phase is water in the paper fibres while the mobile phase is a **developing solvent***.

CE 2011 Paper 2 Q16

* Paper chromatography may be used to provide evidence about inks and pigments used in forgeries. It can also be used to identify the different dyes used in foods.

* Common solvents that are used include pentane, propanone and ethanol.

chromatography 色層法　adsorb 吸附　stationary phase 固定相　mobile phase 流動相
paper chromatography 紙色層法　developing solvent 展開劑

CE 2009 Paper 1 Q7(c)
CE 2009 Paper 2 Q15

AL 1999 Paper 1 Q2(c)

N15 The water in the paper fibres is the stationary phase and the filter paper is the support.

Apply a spot of the solution of a mixture on a base-line on the paper. When the spot is dry, place the paper in the developing solvent (Fig. 54.15a). The solvent rises up the paper and takes the components of the mixture with it.

Each component distributes itself between the water in the paper fibres and the developing solvent. Compounds that are more soluble in the developing solvent travel up more quickly. Thus, the components separate. **N15**

When the solvent front (the leading edge of the solvent) is almost at the top of the paper, take out the paper and dry it. The resulting paper is called a **chromatogram** (Fig. 54.15b).

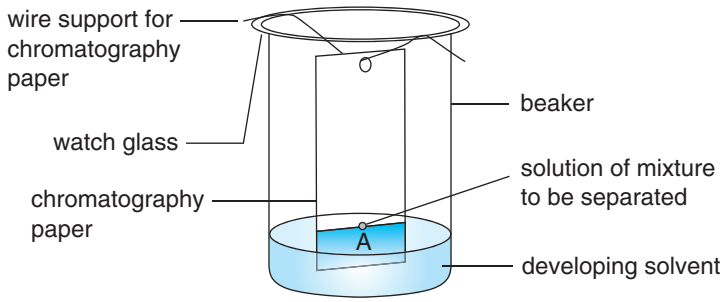


Fig. 54.15a An experimental set-up for paper chromatography

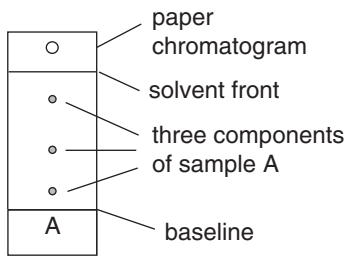


Fig. 54.15b The resulting paper is called a chromatogram

To obtain the separated components, we can cut out the individual spots and dissolve the components using suitable solvents.

Identifying colourless components

N16 Paper chromatography can be used to separate $Cu^{2+}(aq)$ ions from a mixture containing $Cu^{2+}(aq)$ ions and $Cr^{3+}(aq)$ ions. This is because $Cu^{2+}(aq)$ ions and $Cr^{3+}(aq)$ ions have different colours.

With coloured components, the distance moved by a component is easy to see. With colourless components, we need to develop the chromatogram to make the components visible. **N16**

We may spray a *developing agent* such as ninhydrin onto the paper* or place the chromatogram in an atmosphere of iodine vapour. This converts the components into coloured spots on the chromatogram.

* In the separation of amino acids in a mixture, it is usual to locate the spots of separated amino acids by spraying the paper with a chemical called ninhydrin. This chemical forms a purple compound with each amino acid and makes each spot of amino acid visible.

chromatogram 色層譜　　developing agent 顯影劑

Ultraviolet light makes some compounds glow. We may also hold the chromatogram under ultraviolet light and mark the positions of the components (Fig. 54.17).

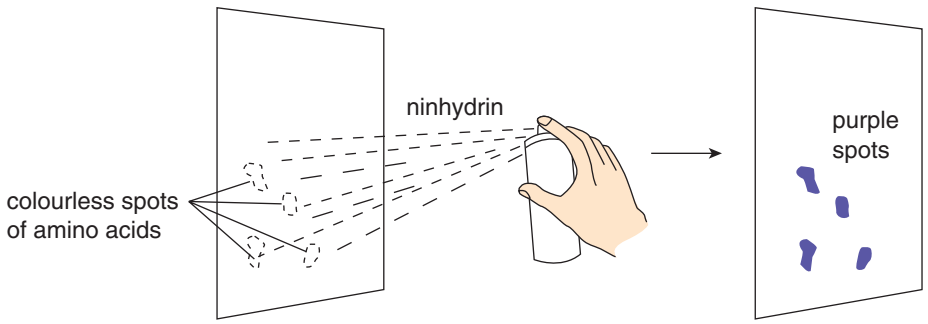


Fig. 54.16 Spraying ninhydrin solution to locate spots of colourless amino acids



Fig. 54.17 Invisible spots can often be seen under ultraviolet light

The retention ratio R_f

The distance a component travels relative to the solvent is a constant as long as the experimental conditions are kept the same. The distance a component travels relative to the solvent is called the **retention ratio** (symbol: R_f).

DSE 2013 Paper 2 Q3(c)(i)
AL 2012 Paper 1 Q8(a)(i)(I)

$$R_f = \frac{\text{distance travelled by the component}}{\text{distance travelled by the solvent}}$$

We can identify a component from its retention ratio.

In the following activity, you are going to separate and identify amino acids by paper chromatography.

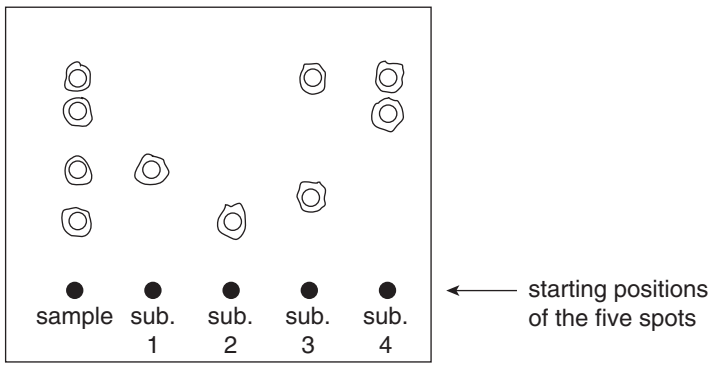
Activity 54.3

Using paper chromatography for the separation and identification of amino acids.

retention ratio 比移值

Checkpoint

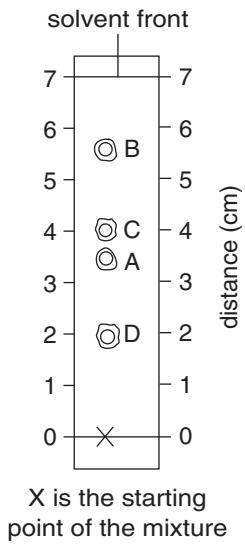
1 The following chromatogram is obtained in the identification of the components in a sample.



Identify the components present in the sample being analyzed.

Substances 1, 2 and 4 are present.

2 A mixture of amino acids, A, B, C and D, was separated by paper chromatography using an appropriate solvent. The chromatogram obtained is shown.



X is the starting point of the mixture

a) Briefly describe the underlying principle for the separation of the amino acids by paper chromatography.

b) The amino acid spots are invisible. Suggest how you can make them visible. Spray a developing agent such as ninhydrin onto the paper. This converts the amino acids into coloured spots on the chromatogram.

c) Calculate the R_f value for B.

R_f value for B $= \frac{5.5 \text{ cm}}{7.0 \text{ cm}}$
$= 0.79$

2 a) Each amino acid distributes itself between the water in the paper fibres and the developing solvent.
Amino acids that are more soluble in the developing solvent travel up more quickly. Thus, the amino acids separate.

54.14 Column chromatography

In **column chromatography**, the stationary phase (an **adsorbent**) is placed in a vertical glass column. Alumina or silica gel is commonly used. The mobile phase, a liquid solvent, is added to the top and runs down through the column.

column chromatography 柱色層法　　adsorbent 吸附劑

Reference Website
Refer to the following website for an animation explaining the principle of column chromatography:
http://chem-ilp.net/labTechniques/ChromatographyAnimation.htm
(accessed May 2015)

Dissolve the mixture in a suitable solvent and carefully load the dissolved sample onto the column. Add more solvent. As the solvent runs down, the components of the mixture separate. The process is called **elution** and the solvent used is called the **eluent** or **eluting solvent**.

AL 2012 Paper 1 Q8(a)(i)(II)

Each component in the mixture has its own equilibrium between adsorption onto the surface of the adsorbent and solubility in the solvent. The better adsorbed ones travel more slowly. The result is a separation into bands containing different components.

Fig. 54.18 shows the separation of three components A, B and C in a mixture. Add eluent to the column until the bands have been eluted and collect each separately.

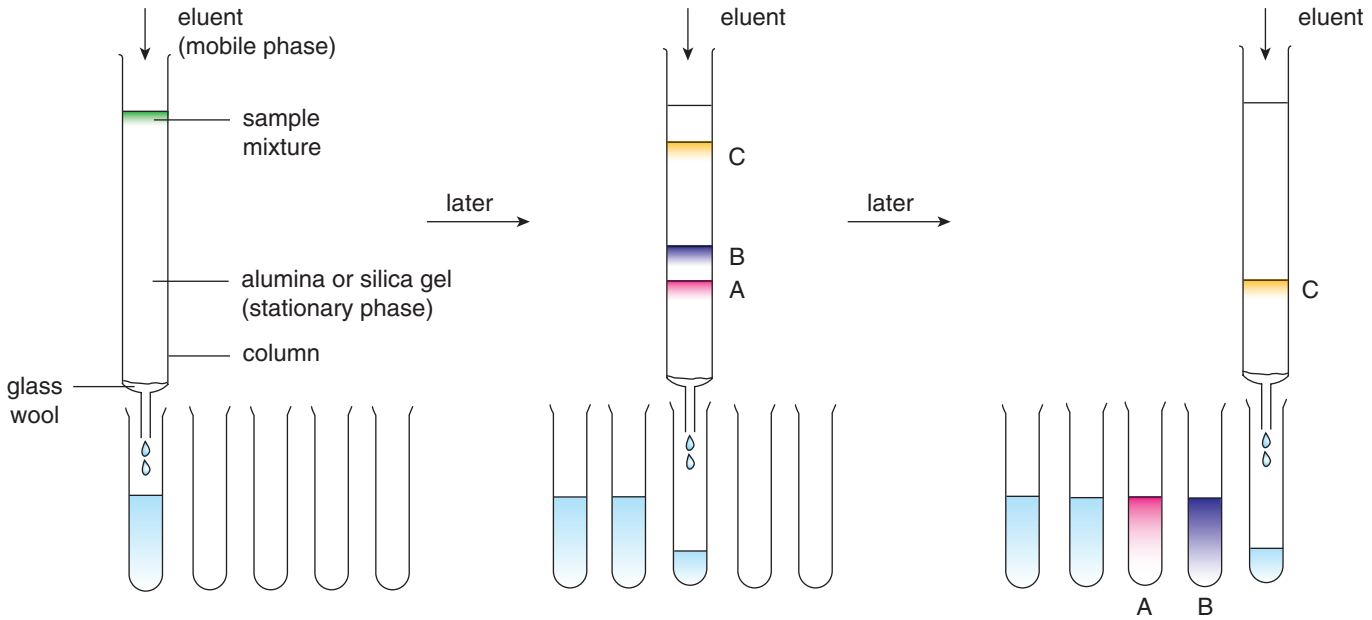


Fig. 54.18 Separation of components A, B and C in a mixture by column chromatography

If the components are coloured, collecting them is easy. If they are colourless, add fluorescent dyes to them and we can see their movement under ultraviolet light.

Reference Website
Refer to the following website for an animation explaining the principle of thin layer chromatography:
http://chem-ilp.net/labTechniques/TLCAnimation.htm
(accessed May 2015)

N17 Refer to page T47.

54.15 Thin layer chromatography

Thin layer chromatography is similar to paper chromatography, except that the stationary phase is a fine layer of alumina or silica gel coated evenly onto a glass plate. N17

elution 洗提　eluent / eluting solvent 洗提液　thin layer chromatography 薄層色層法

N18 Refer to page T48.

* The soaked filter paper ensures that the atmosphere of the jar is saturated with the solvent vapour.

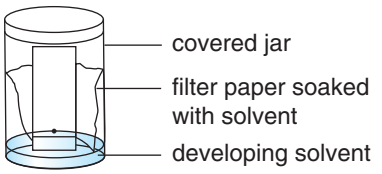


Fig. 54.19 Thin layer chromagtography

Reference Website

Refer to the following website for a video clip on thin layer chromatography:
http://www.youtube.com/watch?v=gzp2S0e9o8s&feature=related
(accessed May 2015)

Apply a spot of the solution of a mixture on one end of the glass plate. After drying, place the glass plate in a jar lined with a piece of filter paper dipping in the solvent*. As the solvent rises up by capillary action, the components of the mixture are carried up and separated. When the solvent is almost at the top of the plate, take the plate out and allow it to dry. N18

We can see the separation of coloured components easily. But if they are colourless, we can put the dry, already developed glass plate in a beaker containing some iodine crystals. The iodine vapour slowly oxidizes most organic compounds, making the components shown up as brown spots on the plate.

We can obtain the separated components by scraping off the individual spots and dissolving them in a suitable solvent.

Table 54.5 shows the applications of each type of chromatography discussed above.

Table 54.5 Applications of some types of chromatographic methods

Chromatographic method	Stationary phase	Mobile phase	Separation	Identification	Testing purity
Paper	water in the paper fibres	liquid solvent		✔	✔
Column	adsorbent in vertical glass column	liquid solvent	✔		
Thin layer	fine layer of alumina or silica gel coated onto a glass plate	liquid solvent		✔	✔

54.16 Tests for purity of a product

After the separation and purification of a product, the next step is to check the purity of the product.

For a solid product, measuring its melting point helps to check whether it is pure. A pure product has a very sharp melting point over a very narrow temperature range. Another useful technique for checking the purity of a solid product is thin layer chromatography. A pure compound produces only one spot on the chromatogram.

We can use the boiling point of a liquid product to check its purity.

Reference Website
Refer to the following website for a video clip on determining the melting point of a solid:
http://chem-ilp.net/labTechniques/MeltingPointsVideo.htm
(accessed May 2015)

Determining the melting point of a solid product

ASL 2010 Paper 1 Q10(c)
AL 2010 Paper 1 Q8(f)

A pure solid usually has a very sharp melting point and melts over a very narrow temperature range (approximately 1 – 2 °C). Presence of even 1% impurity can lower the melting point and widen the range to several degrees. Therefore measuring the melting point of a solid product is a good way of checking its purity. N19

We can use the set-up shown in Fig. 54.20 to determine the melting point of a solid product. First place some of the dry solids in a glass capillary tube*. Then tie the tube to a thermometer with a rubber band. The solids in the tube should be level with the bulb of the thermometer. Place this set-up together with a stirrer in an oil bath* and heat gently with stirring.

Fig. 54.21 shows an electrical melting point apparatus.

* The glass capillary tube should be sealed at the bottom.

* The boiling point of the oil should be higher than the melting point of the solid product.

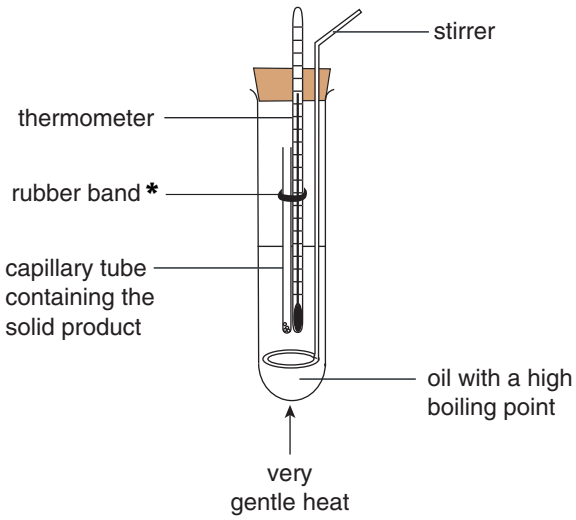


Fig. 54.20 An experimental set-up for determining the melting point of a solid product

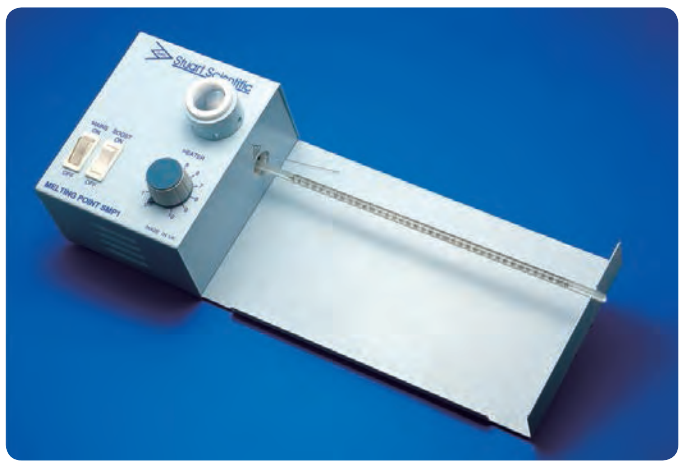


Fig. 54.21 An electrical melting point apparatus

* Do not allow the rubber band to come into contact with the oil in the bath.

The melting point range of the solid is the temperature at which we first observe the solid to melt until the melting is complete.

Determining the boiling point of a liquid product

We can use the boiling point of a liquid product to identify it or to check its purity. It is difficult to find a sharp boiling point. Usually, we can only obtain a boiling point range of a 2 – 3 °C accuracy.

Use the set-up shown in Fig. 54.22 to determine the boiling point of a liquid product.

N19 Aldehydes and ketones react with 2,4-dinitrophenylhydrazine to form 2,4-dinitrophenylhydrazones. We can determine the melting point of the 2,4-dinitrophenylhydrazone formed from an aldehyde or a ketone and compare it with melting points stated in data books to identify the aldehyde or ketone.

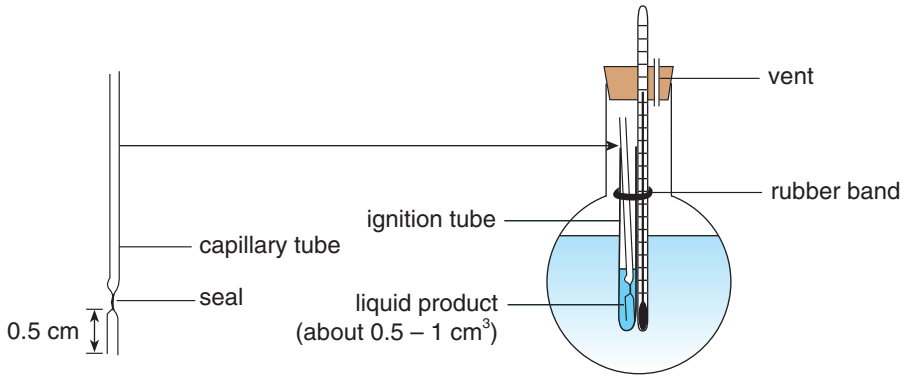


Fig. 54.22 An experimental set-up for determining the boiling point of a liquid product

Place a little of the liquid product in a clean ignition tube. Insert a capillary tube sealed at 0.5 cm from one end into the liquid. Tie the ignition tube to the thermometer and place this set-up together in an oil bath.

Heat the bath slowly. As the temperature is increased, evolution of bubbles from the end of the capillary tube begins. When the bubbles are coming out at a very rapid pace, stop the heating.

Carefully observe the capillary tube. As the temperature decreases, the bubbles will slow down. The boiling point of the liquid product is reached when the bubbles stop.

We can also use the simple distillation set-up shown in Fig. 54.23 to determine the boiling point of a liquid product.

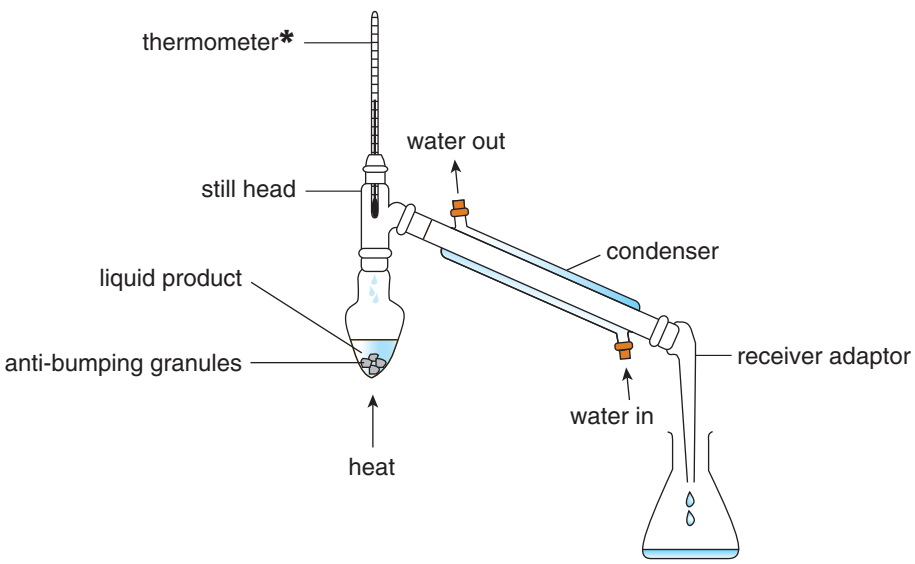


Fig. 54.23 Determining the boiling point of a liquid product using a simple distillation set-up

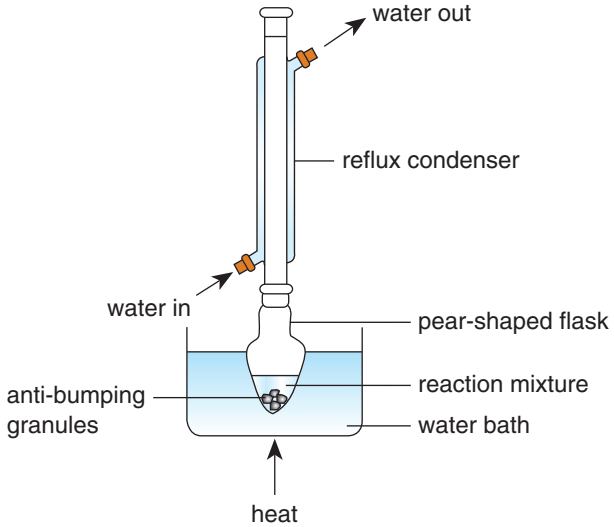
* The bulb of the thermometer is placed near to the side-arm of the still head connected to the distillation flask. This is to record the temperature of the vapour which is distilled over, i.e. the boiling point of the liquid product.

AL 2005 Paper 1 Q7(a)(i)

Checkpoint

Refer to page T48.

1 In an experiment to prepare cyclohexene from cyclohexanol, a mixture of cyclohexanol and phosphoric acid is heated in a water bath at 70 °C for 15 minutes as shown below.



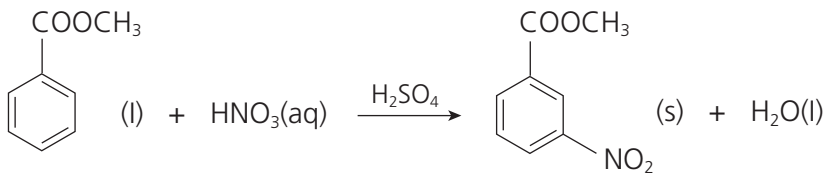
a) Suggest how to separate the crude cyclohexene from the reaction mixture.

b) Name TWO impurities present in the crude cyclohexene.

c) Suggest how to purify the crude cyclohexene.

(Density of cyclohexene = 0.811 g cm^{-3}; boiling point of cyclohexene = 83 °C)

2 Consider the preparation of methyl 3-nitrobenzoate from methyl benzoate and concentrated nitric acid.



methyl benzoate — methyl 3-nitrobenzoate

A mixture of methyl benzoate, concentrated nitric acid and concentrated sulphuric acid is allowed to stand at room temperature for 10 minutes.

Suggest how to

a) separate methyl 3-nitrobenzoate from the reaction mixture;

b) purify the methyl 3-nitrobenzoate;

c) determine the purity of the product.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

Potable water

The sources of potable water include reservoirs, rivers, lakes, natural springs and wells. As water moves across the Earth's surface or through rocks and soil, it dissolves minerals, gases and various organic materials. These substances are carried into sources of drinking water.

It is the job of chemists employed by water authorities to monitor and manage our water supplies. In the case of potable water, they must ensure that the levels of microorganisms, salts, heavy metals, suspended matter, coloured matter and organic substances are within acceptable ranges.



Fig. 54.24 Drinking water from the tap



Fig. 54.25 Sedimentation tanks at a water purification plant



Fig. 54.26 Tanks of chlorine at a waterworks. What is the use of chlorine here?

Purification of drinking water

The figure below shows the steps involved in the purification of our drinking water.

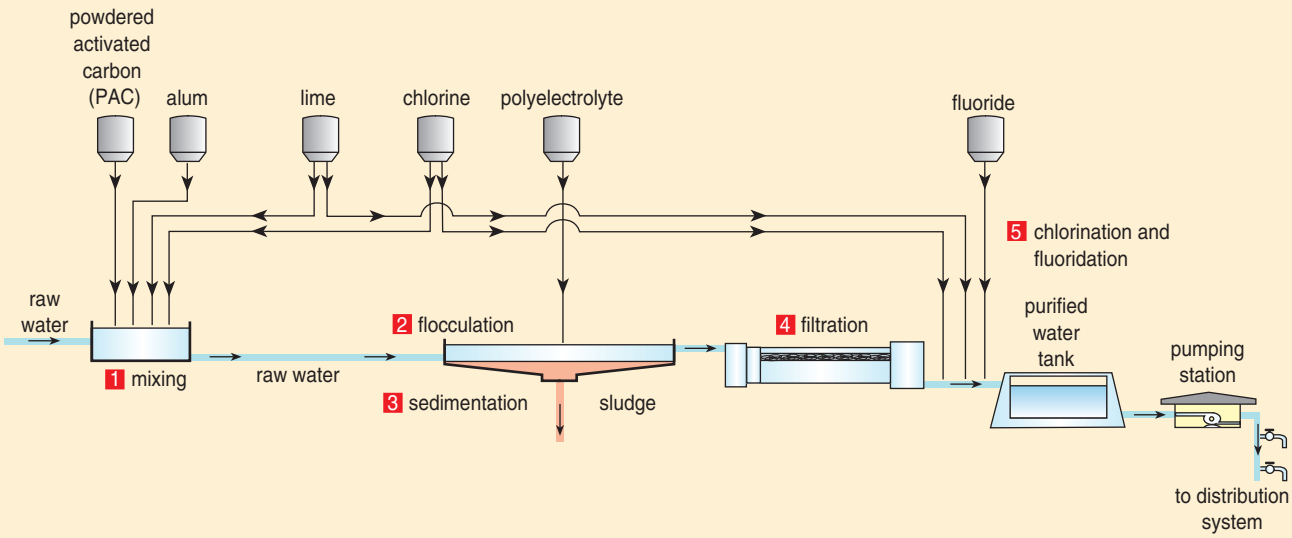


Fig. 54.27 Steps in the purification of drinking water

Step 1 Mixing

The raw water is mixed with the following chemicals:

- lime — to pre-condition the raw water prior to the addition of alum;
- chlorine — to suppress the *proliferation* of algae;
- alum — to *flocculate* suspended particles;
- powdered activated carbon — to remove tastes and odours when required.

Step 2 Flocculation

Water in rivers and reservoirs contains small suspended particles. These can be made to precipitate by a process called flocculation. The alum (aluminium sulphate) added to the water produces a floc of aluminium hydroxide. This floc traps suspended particles.

Step 3 Sedimentation

The treated water is allowed to stand so that the floc settles to the bottom to form sludge.

Step 4 Filtration

The water is then passed through a filter containing sand and anthracite coal to remove the more finely divided suspended particles.

Step 5 Chlorination and fluoridation

Chlorine is added into the filtered water for disinfection and fluoride added for dental protection.

Lime is added to achieve the desired pH. Adjustment of pH is important as this reduces the corrosion of water pipes.

Water Distillation

Distillation is another method of water purification. Distillation is an option where the raw water contains either an objectionable odour, a bad taste or high concentrations of dissolved minerals. The method is also suitable for coastal areas where sea water is abundant but potable water is not available.

proliferation 迅速繁殖　　flocculate 絮凝

Making water safe for drinking

There is often a need to treat water in remote or rural locations to make it safe for drinking purposes. A common water treatment method used by hikers and campers is chemical disinfection with iodine. Tincture of iodine (found in some first aid kits) is one possible source of iodine. The unpleasant taste of iodine can be countered by adding ascorbic acid (vitamin C) to the treated water.

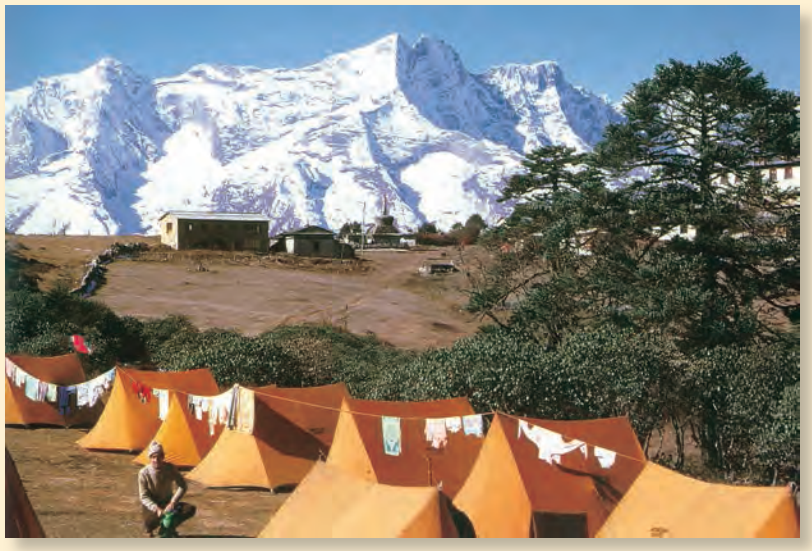


Fig. 54.28 Campers using water from a river or lake can make it safe for drinking by disinfection with iodine

Questions

1 Discuss the pros and cons of using distillation for water purification.

2 Besides treatment with iodine, suggest other water treatment methods of making water in remote or rural locations safe for drinking.

You may refer to the following website for information:
http://wwwnc.cdc.gov/travel/page/water-disinfection

STSE Connections

1 Pros of using distillation for water purification
- Distillation is often used as the preferred water purification method in developing nations or areas where the risk of waterborne disease is high, due to its unique capabilities to remove bacteria and viruses from drinking water.
- Distilled water will not cause scales in water-using appliances like coffee maker, iron or clothes steamer.

Cons of using distillation for water purification
- Distillation is expensive.
- Distillation removes every possible impurity from water, including minerals and salts. The human body is incapable of surviving on distilled water for a prolonged period of time, as it does not contain the much needed composition of minerals. These minerals are very much needed by the body and a deficiency is bound to cause some ailments.
- The content of salts and minerals also give water a peculiar and welcoming 'taste'.

2 Four other water treatment methods:
- boiling;
- portable filters providing various degrees of protection;
- ultraviolet light;
- solar radiation.

References:
http://www.allaboutwater.org/distillation.html
http://www.enviroalternatives.com/watermethods.html

Summary

1 The following table summarizes results of chemical tests for the alkene functional group.

Test	Alkene functional group –C=C–
Aqueous bromine	the yellow-brown aqueous bromine becomes colourless rapidly
Cold acidified dilute aqueous solution of potassium permanganate	the purple solution becomes colourless rapidly

2 The following table summarizes results of chemical tests for functional groups containing oxygen.

Test \ Functional group	Alcohols —O—H	Aldehydes —C(=O)—H	Ketones —C(=O)—	Carboxylic acids —C(=O)—OH
Warm with $K_2Cr_2O_7$ / H_3O^+	1° and 2° alcohols* — clear orange solution turns green almost immediately 3° alcohols* — no observable change	clear orange solution turns green	—	—
Treatment with 2,4-dinitrophenylhydrazine	—	a yellow to red precipitate forms	a yellow to red precipitate forms	—
Warm with Tollens' reagent (an aqueous solution of silver nitrate in ammonia)	—	silver mirror forms on the wall of reaction vessel	—	—

Extension

* 1°, 2° and 3° alcohols can be distinguished by treating with the Lucas reagent (anhydrous $ZnCl_2$ in concentrated HCl):

- 1° alcohols — cloudiness appears in hours
- 2° alcohols — cloudiness appears in 5 – 10 minutes
- 3° alcohols — cloudiness appears in 1 minute

Continued on next page

Test \ Functional group	Alcohols — O — H	Aldehydes O ‖ — C — H	Ketones O ‖ — C —	Carboxylic acids O ‖ — C — OH
Extension Iodoform test (warm with iodine in an aqueous solution of sodium hydroxide)	alcohols containing the OH | — C — CH_3 | H group gives a bright yellow precipitate	aldehydes and ketones with the O ‖ — C — CH_3 group gives a bright yellow precipitate		—
Treatment with aqueous sodium hydrogencarbonate solution	—	—	—	a colourless gas (CO_2) given off
Warm with ethanol in the presence of concentrated sulphuric acid, followed by pouring the mixture into an aqueous solution of sodium carbonate	—	—	—	a sweet, fruity-smelling ester can be detected

Summary

3 The following table summarizes common separation and purification methods for liquid and solid products.

Type of product	Method	Remark
Liquid product	distillation	• If other substances in a reaction mixture are non-volatile, separate the crude liquid product by distillation.
	fractional distillation	• Separate a mixture of miscible liquids by fractional distillation, providing that the boiling points of the liquids are not too close (larger than 10 °C).
	liquid-liquid extraction	• In the preparation of carbon compounds, liquid-liquid extraction may be used to extract the liquid product from the aqueous reaction mixture. • The extraction method is based on the principal that carbon compound products are usually more soluble in organic solvents than in water.
Solid product	re-crystallization	• The method is often used to purify crude solid products. • The solid product should dissolve readily in a hot solvent but is much less soluble when the solvent cools down.
	chromotography	• The method requires two different phases — a stationary phase and a mobile phase. • The basic principle of chromatography is that components in a mixture have different tendencies to adsorb onto a surface or dissolve in a solvent.

4 As long as the same solvent and type of chromotographic paper are used, we can identify a component in a mixture from its retention ratio R_f.

$$R_f = \frac{\text{distance travelled by the component}}{\text{distance travelled by the solvent}}$$

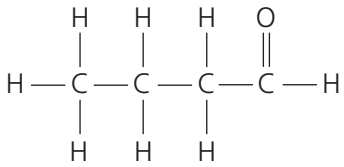
5 The purity of a solid product may be checked by its melting point and that of a liquid product by its boiling point.

Exercise

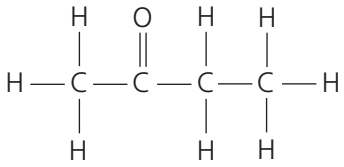
Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Compounds A and B are isomers.



A



B

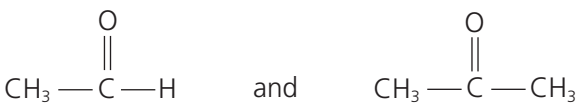
Complete the table with the expected observations when A and B are treated separately with the following reagents.

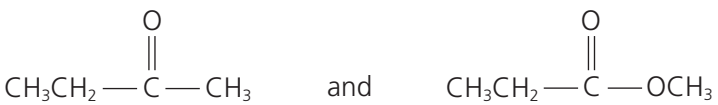
Reagent	Observation with A	Observation with B
2,4-dinitrophenylhydrazine	a yellow to red precipitate forms	a yellow to red precipitate forms
Warm with Tollens' reagent	silver mirror forms on the wall of reaction vessel	no observable change
Warm with iodine and an aqueous solution of sodium hydroxide	no observable change	a bright yellow precipitate forms

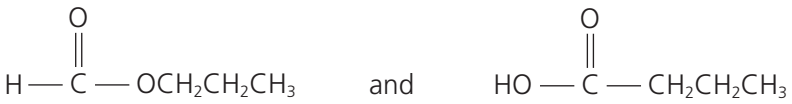
Extension

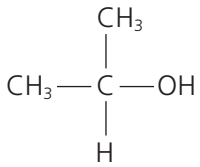
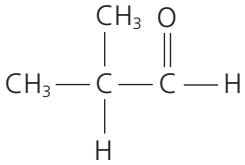
2 Pentan-1-ol and pentan-2-one can be distinguished from each other by the use of potassium dichromate. State the conditions required and describe the chemistry involved.
Refer to page T49.

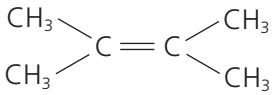
3 For each of the following pairs of chemicals, suggest a chemical test to distinguish them. State also the expected observations and write relevant equations.
Refer to page T49.

a) 

b) 

c) 

d)  and 

e)  and 

4 Below is some information concerning two black ink samples:

Sample A — contains only one black dye
Sample B — contains 3 dyes: yellow, green and blue

4 a) Paper chromatography
b) Sample A – one black spot
Sample B – three colour spots (yellow, green and blue)

a) What process would you use to distinguish the two samples?

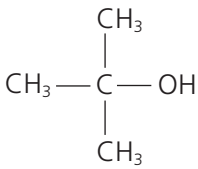
b) Using diagrams, show the results you would expect on subjecting each sample to the process.

5 Describe, with the help of a labelled diagram, how you can determine the melting point of a solid sample.
Refer to page T51.

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

6 Alcohol X has the following structure:
Refer to page T51.

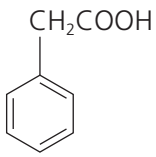


a) Draw the structure of three structural isomers of X, all of which are alcohols.

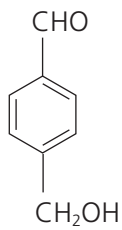
Extension

b) Describe how a solution of anhydrous zinc chloride in concentrated hydrochloric acid can be used to distinguish X from the three structural isomers.

7 X and Y are isomers with the molecular formula $C_8H_8O_2$.
Refer to page T52.



X



Y

a) Complete the table with the expected observations when X and Y are separately warmed gently with the following reagents.

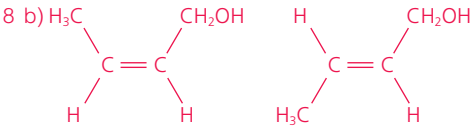
Reagent	Observation with X	Observation with Y
Aqueous solution of sodium hydrogencarbonate		
2,4-dinitrophenylhydrazine		
Acidified aqueous solution of potassium dichromate		

b) Both X and Y can form esters.

i) Draw the structural formula of the ester which forms when X reacts with ethanol.

ii) Draw the structural formula of the ester which forms when Y reacts with ethanoic acid.

8 In order to establish the functional groups present in an acyclic compound X, C_4H_8O, three chemical tests were carried out. The test results are listed in the table below.

8 b) 

a) Complete the table by giving the deduction for each test.

Test reagent	Test result	Deduction
$K_2Cr_2O_7$ / H_3O^+	positive	functional group(s) that may be present: —OH, —CHO
Tollens' reagent	negative	functional group(s) absent: —CHO
Aqueous bromine	positive	functional group(s) that may be present: C=C, C≡C

b) Compound X can be oxidized to a carboxylic acid which has a *cis-trans* isomer.

Based on all of the above information, draw two possible structures of X.

9 Aromatic compound X has the molecular formula $C_8H_{10}O$.

Refer to page T52.

Consider the following test results:

Test 1 Oxidation of X by dichromate ion yields Y (C_8H_8O). Y gives a negative result when warmed with Tollens' reagent.

Test 2 X reacts with ethanoic acid in the presence of concentrated sulphuric acid to give a sweet-smelling liquid Z.

a) With reference to *Test 1*, deduce the structure of Y.

b) i) With reference to *Test 2*, deduce the functional group that X may have.

ii) Deduce the structure of X.

c) Draw the structure of Z.

10 This question is about butanal, $CH_3CH_2CH_2CHO$, and several related compounds.
Refer to page T53.

a) What would you expect to see when butanal is treated with 2,4-dinitrophenylhydrazine.

b) X, Y and Z are three isomers of butanal. The following tests are carried out to study them.

	Test	X	Y	Z
I	Treat with 2,4-dinitrophenylhydrazine	an orange precipitate is formed	an orange precipitate is formed	no observable change
II	Warm with Tollens' reagent	no observable change	a silver mirror forms	no observable change
III	Shake with an acidified aqueous solution of potassium dichromate	no observable change	the orange solution turns green	the orange solution turns green

Suggest the structural formulae of X, Y and Z. Explain your reasoning.

11 A is an aromatic compound with the molecular formula C_9H_8O. It gives positive results when treated with Tollens' reagent. It also decolourizes aqueous bromine readily.
Refer to page T53.

a) Deduce all functional groups present in A.

b) State a type of isomerism that A can exhibit. Illustrate your answer with the appropriate structures.

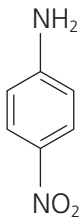
12 You are provided with four unlabelled bottles each containing one of the following colourless liquids:

1-bromopropane, butan-1-amine, cyclohexene, propanone

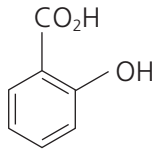
Outline a scheme of tests to distinguish the four liquids from one another.

(HKASLE 2005)

13 A student carried out an experiment to extract 4-nitrophenylamine and 2-hydroxybenzoic acid from a solution of the two compounds in ethoxyethane. The solution contained 1.5 g each of the compounds in 20 cm^3 of ethoxyethane.



4-nitrophenylamine



2-hydroxybenzoic acid

The student shook the solution with 12 cm^3 of 0.5 M $NaHCO_3(aq)$ in a separating funnel. The mixture in the separating funnel was allowed to settle, and the aqueous layer was then separated from the organic layer. Evaporation of the aqueous layer gave a solid residue, which was water-soluble and had a melting point higher than 200 °C. Evaporation of the organic layer gave another solid residue, which melted over a wide temperature range of 140 °C to 160 °C.

a) Why was $NaHCO_3(aq)$ used in the extraction?

b) While the student was shaking the separating funnel, the stopper of the funnel popped out, spilling part of the mixture on the bench. With the help of equation(s), explain why this happens.

c) Was the solid residue obtained from the organic layer a pure compound? Support your answer by stoichiometric calculation.

d) State TWO reasons why ethoxyethane is commonly used to extract organic compounds from an aqueous solution.

e) State ONE safety precaution necessary in the evaporation of ethoxyethane.

(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0)

(HKALE 2003)

14 In an experiment to prepare 1-bromobutane, a mixture of butan-1-ol, potassium bromide and concentrated sulphuric acid is heated under reflux for 30 minutes.

a) Suggest how to isolate 1-bromobutane from the reaction mixture.

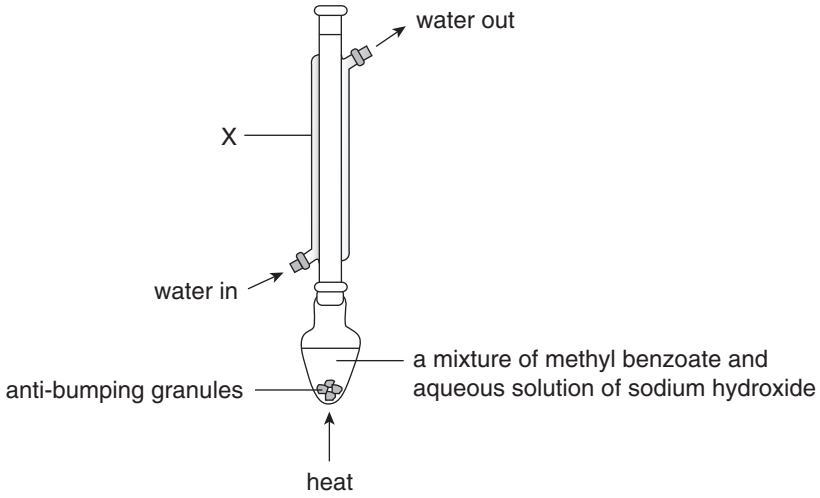
Isolate the crude 1-bromobutane from the reaction mixture by distillation.

b) Describe how you would purify the crude 1-bromobutane.

The distillate collected has an organic and an aqueous layer. Use a teat pipette to remove the aqueous upper layer.
Transfer the organic layer to a separating funnel and add concentrated hydrochloric acid. Stopper and shake well. This removes the unreacted butan-1-ol as it is soluble in the acid.
Allow the two layers to separate. Run the organic layer into a conical flask.
Shake the crude 1-bromobutane with sodium hydrogencarbonate solution in another separating funnel. This removes the excess hydrochloric acid and other acid impurities.
Allow the layers to separate and run the organic layer into a conical flask.
Add anhydrous sodium sulphate to dry the 1-bromobutane. Swirl the mixture until the liquid is clear.
Decant the dried product into a pear-shaped flask. Carry out fractional distillation. Collect the fraction with the boiling point range of 1-bromobutane. This removes any remaining organic impurities.

15 In an experiment to prepare benzoic acid, a mixture of methyl benzoate and aqueous solution of sodium hydroxide is heated using the experimental set-up shown below.

Refer to page T54.



a) Name apparatus X.

b) Write the chemical equation for the reaction involved.

c) Suggest how you can obtain a crude sample of benzoic acid from the resulting mixture.

d) The crude sample of benzoic acid can be purified by re-crystallization, using water as the solvent. Describe how you would carry out this re-crystallization.

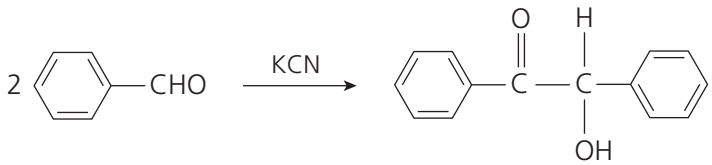
e) How can you check that the purification is successful?

16 a) A sample of N-phenylethanamide is contaminated with some coloured impurities. Outline the procedures for the purification of this sample by re-crystallization from water.

b) Draw a labelled diagram for the assembly of apparatus used in simple distillation.

(HKALE 2004)

17 14.6 g of benzoin was obtained by heating 15.9 g of benzaldehyde with potassium cyanide in a solvent-free condition.



benzoin (m.p. 137 °C)

a) Calculate the percentage yield of benzoin from benzaldehyde.

b) Devise an experimental procedure to remove potassium cyanide from the crude product.

c) The benzoin obtained can be purified by re-crystallization. Suggest THREE criteria for an appropriate solvent for the re-crystallization.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

(HKALE 2007)

18 Scientists test three orange drinks to see if they contain a banned colour additive. They use the technique of paper chromatography. This is a description of what the scientists do.

Refer to page T55.

Step 1 They draw a pencil line 2 cm from the bottom of a piece of absorbent paper.

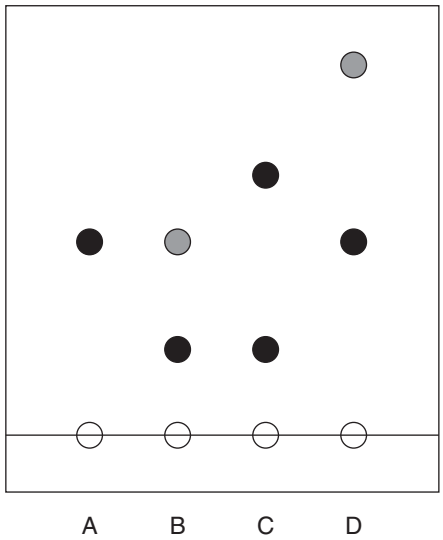
Step 2 They put a spot of the banned colour additive and each of three orange drinks on the pencil line. They use a pencil to label the banned colour additive A and the orange drink spots B, C and D.

Step 3 They place the bottom of the paper into 1 cm depth of water in a beaker.

Step 4 They cover the beaker with a sheet of glass and leave it until the water has soaked almost to the top of the paper.

Step 5 They remove the paper from the beaker and leave it to dry.

This is what the paper looks like at the end of this process.



a) Explain how the colours in the orange drink samples became separated on the chromatography paper.

b) The scientists decide that only one of the three orange drinks contains the banned colour additive. Explain how their results show this.

c) Scientists also test a permanent felt tip pen ink to see if it contains a prohibited dye. For this test they use thin layer chromatography.

i) Describe one similarity and one difference between thin layer chromatography and paper chromatography.

ii) Suggest why the scientists use thin layer chromatography instead of paper chromatography for this test.

iii) The scientists find that one spot from the felt tip pen ink and the spot from the prohibited dye look similar and seem to have travelled the same distance on the chromatogram.

The scientists measure the distance travelled by each of these two spots, and the distance travelled by the solvent front above each spot.

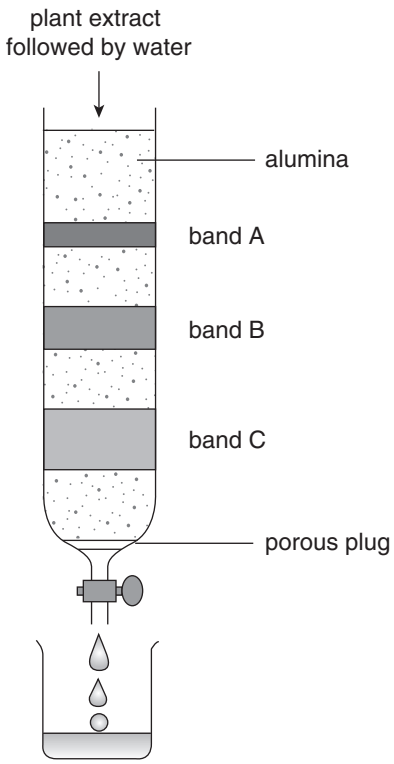
	Distance travelled in mm	
	Solvent front	**Spot**
Felt tip pen ink	96	44
Prohibited dye	98	45

Calculate the R_f value for the felt tip pen ink and for the prohibited dye. You are advised to show how you work out your answer. 0.46; 0.46

iv) Suggest why the scientists might want to calculate R_f values.

(OCR GCSE, Chemistry A, Specimen Paper)

19 A glass column packed with alumina (Al_2O_3) is used to separate the components in a plant extract by chromatography. A solution of the plant extract is placed at the top of the alumina. Water is further added periodically. The components separated as three coloured bands, as shown in the diagram below.
Refer to page T55.



a) In which one of the three bands, A, B or C, are the components most strongly adsorbed to the stationary phase? Explain your answer.

b) In another experiment, the components in the plant extract are separated by thin layer chromatography, using a glass sheet coated with alumina and water as the mobile phase. Which one of the three bands, A, B or C, contains the component that would have the greatest R_f value? Explain your answer.

20 a) Draw a labelled diagram for the assembly of apparatus that you would use to determine the boiling point of propan-2-ol.

b) Suggest ONE hazard warning label which should be displayed on a bottle of propan-2-ol.

(HKALE 2005)

21 Read the procedure of each of the following experiments. Identify all the hazards involved. Present the risk assessment in table form as shown below.

Refer to page T56.

Experiment title:			
Hazardous chemical, procedure or equipment involved	Nature of hazard	Safety precautions	Source of information

Experiment A — Testing methanal with Tollens' reagent

1 Place 3 cm^3 of $AgNO_3(aq)$ into a clean test tube and add dilute $NH_3(aq)$ drop by drop until the precipitate of silver oxide just dissolves. The clear solution contains $[Ag(NH_3)_2]^+$ ions.

2 Add 10 drops of methanal and warm the resulting mixture in a beaker of hot water.

Experiment B — Preparing 2-chloro-2-methylpropane

1 Pour 9 cm^3 of methylpropan-2-ol into a separating funnel.

2 Add 20 cm^3 of concentrated hydrochloric acid, about 3 cm^3 at a time. After each addition, invert the separating funnel several times to mix the contents.

3 Leave the separating funnel and the contents in a fume cupboard for about 20 minutes. Shake it gently at intervals.

4 Allow the layers in the separating funnel to separate. Run off and discard the lower aqueous layer.

22 Some experiments are carried out on a compound X (C_3H_6O) and the results are shown below.

Test	Procedure	Observation	Deduction
1	Add 2 cm^3 of X to 10 cm^3 of water.	X is soluble in water.	X may form hydrogen bonds with water.
2	Add 1 cm^3 of dilute sulphuric acid to 2 cm^3 of aqueous solution of potassium dichromate. Add 10 drops of X and warm gently.	The dichromate solution changes from orange to green.	The dichromate ions are reduced to chromium(III) ions. X is either a primary / secondary alcohol, or an aldehyde.
3	Mix 2 cm^3 of X with 2,4-dinitrophenylhydrazine	An orange precipitate forms.	X is either an aldehyde or a ketone.

a) Complete the 'Deduction' column based on the results of the tests.

b) Deduce, with explanation, the functional group(s) present in X.

Based on the results of tests 2 and 3, X is probably an aldehyde. Thus, X contains $-\overset{\overset{O}{\|}}{C}-H$ group.

c) Draw the structure of X.

$CH_3CH_2-\overset{\overset{O}{\|}}{C}-H$